O ‘ QUVQO ‘ LLANMA · 2 0 2 6 - YILKUZ

Math I

1–4 haftalar

Har bir kurs fayli va tayinlangan darslik bo‘limlaridagi barcha savollar va formulalar — son sistemalari, yozuv, kompleks sonlar; funksiyalar; limitlar va limit qonunlari, standart limitlar, aniqlanmagan holdalar, asimptotika va Landau belgilari — har bir masala to‘liq bosqichma-bosqich yechim bilan. Mustaqil o‘rganish va Test 1 ga tayyorgarlik uchun tuzilgan

Toshkent texnologiya universiteti · 2026-yil Kuz

Manbalar: kurs konspektlari, darslar, *ISA*, amaliy mashqlar; OpenStax Calculus 1

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1 Kurs xaritasi va hujjatdan qanday foydalanish

Hujjat bu 2026-yil kuzida birinchi semestrda bo'ladigan Math I kursining birinchi to'rt haftasiga oid to'liq mustaqbil qo'llanmadir. U kurs course-files/ papkasidagi har bir tegishli manbadan qo'lda tuzilgan: haftalik konsppektlar (Notes), ma'ruzalar prezentatsiyalari, amaliy mashqlar varaqchalari, mustaqbil ish uch vazifalar (ISA) va tayinlangan OpenStax Calculus Volume 1 darslik bo'limlari. Masalalar so'zma-so'z takrorlangan; yechimlar to'liq, bosqichma-bosqich, tekshirishlar bilan.

Hafta	Math I
1	Sonli to'plamlar $N \subseteq Z \subseteq Q \subseteq R \subseteq C$; pozitsion tizimlar; ilmiy va muhandislik yozuvi; kompleks sonlar
2	Funksiyalar: sinflar, aniqlanish sohasi/qiyofati, simmetriya, kompozitsiya, teskari funksiyalar; eksponensial va logarifmik funksiyalar (§1.2, §1.5)
3	Limitlar: intuitiv tushuncha, bir tomonlama va cheksiz limitlar, asimptotalar, aniq $\varepsilon - \delta$ ta'rifi (§2.2, §2.5)
4	Limitlar II: limitlar qonunlari, standart limitlar, noaniqliklar, ikki militsioner teoremasi, asimptotik ekvivalentlik, Landau $Ovao$ belgilari (§2.3)

Yozuv. Darajalar va indekslar ^{} va _ orqali yoziladi: $x^2, a_{11}, f^{(-1)}$. To'plamlar standart yoziladi: $N \subseteq Z \subseteq Q \subseteq R \subseteq C$. Hosila $f'(x)$ yoki dy/dx bilan belgilanadi.

2 Hafta 1 — Sonli tizimlar, yozuv, kompleks sonlar (Math I)

2.1 Math I: 1-hafta formulalari katalogi

2.1.1 Sonlar to'plamlari va pozitsion sistemalar

Standart sonlar to'plamlarining ichma-ich joylashuvi kursning birinchi struktura fakti bo'lib, keyingi har bir tasniflash mashqi unga tayanadi. Ratsional sonlar — aynan butun sonlarning kasrlari, irratsional sonlar esa ratsional bo'lmagan haqiqiy sonlardir.

$$N \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}, \quad \mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\} \text{ (irratsionallar)} \tag{1}$$

$$q \in \mathbb{Q} \Leftrightarrow q = \frac{a}{b},\ a \in \mathbb{Z},\ b \in \mathbb{Z},\ b \neq 0, \quad d(a,b) = |a-b| \ (\mathbb{R} \text{ dagi masofa}) \tag{2}$$

Asosi b bo'lgan pozitsion sanoq sistemasi sonni asosning darajalari orqali yozadi; bir sonning ko'p yozuvlari bo'lishi mumkin, lekin uning qiymati yozuvdan mustaqil. Asoslarni o'zgartirish quyidagi yoyilmaga (va teskari yo'nalishda ketma-ket bo'lishga) tayanadi.

$$(a_n a_{n-1} \dots a_1 a_0)_b = \sum_{k=0}^{n} a_k b^k, \quad 0 \le a_k \le b-1 \tag{3}$$

$$13_{10} = 1101_2 = 15_8 = D_{16}, \quad A = 10,\ B = 11,\ \dots,\ F = 15 \text{ (o'n oltilik raqamlar)} \tag{4}$$

Rim raqamlari nopozitsion: $I = 1, V = 5, X = 10, L = 50, C = 100$, shuning uchun $13 = XIII$; bunday sistemalarda arifmetika amaliy emas, kompyuterlar esa ikkilik sistemadan foydalanadi: elektronika ikki holatni (past/baland, o'chirilgan/yoniq, 0/1) ishonchli ajratadi.

2.1.2 Ilmiy va muhandislik yozuv

Ilmiy yozuv sonning kattaligini nuqtadan oldingi bitta noldan farqli raqamli koeffitsient va o'nlik daraja orqali ajratadi; muhandislik yozuvida esa daraja uchlikkar ko'paytmasi bo'lishi shart, shunda daraja SI prefiksi bilan bog'lanadi.

$$\text{ilmiy: } z = a \times 10^n, \quad 1 \le |a| < 10,\ n \in \mathbb{Z} \tag{5}$$

$$\text{muhandislik: } z = a \times 10^{3k}, \quad 1 \le |a| < 1000,\ k \in \mathbb{Z} \tag{6}$$

$$(a \times 10^m)(b \times 10^n) = ab \times 10^{m+n}, \quad \frac{a \times 10^m}{b \times 10^n} = \frac{a}{b} \times 10^{m-n} \tag{7}$$

$$(a \times 10^n)^m = a^m \times 10^{nm}, \quad \text{qo'sh./ayir.: avval umumiy darajaga keltirish} \tag{8}$$

Butun semestr davomida ishlatiladigan SI prefikslari:

Qo'shimcha	Belgi	Daraja
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
milli	m	10^{-3}
mikro	µ	10^{-6}
nano	n	10^{-9}
piko	p	10^{-12}
femto	f	10^{-15}

Registr muhim: 5 mW $= 5 \times 10^{-3} W$, lekin $5 MW = 5 \times 10^{6} W$. Kurs bo'ylab ishlatiladigan ma'lumot qiymatlari:

$$c = 2.99792458 \times 10^8 \text{ m/s}; \quad e = 1.602176634 \times 10^{-19} \text{ C}; \quad N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1};$$

$$h = 6.62607015 \times 10^{-34} \text{ J·s}; \quad 101325 \text{ Pa} = 101.325 \text{ kPa}.$$

2.1.3 Kompleks sonlar

1-hafta kompleks sonlar bilan yakunlanadi; ular $x^2+1=0$ tenglamasining haqiqiy yechimi yo'qligi sababli paydo bo'ladi. Muhayyar birlik, tenglik qoidasi, to'rt arifmetik amal va qo'shma son — asosiy ish vositalari.

$$i=\sqrt{-1},\ i^2=-1,\ z=a+bi,\ \operatorname{Re}(z)=a,\ \operatorname{Im}(z)=b \tag{9}$$

$$a+bi=c+di \Leftrightarrow a=c \text{ va } b=d, \quad i^{n+4}=i^n\ (i^0=1,\ i^1=i,\ i^2=-1,\ i^3=-i) \tag{10}$$

$$z_1+z_2=(a+c)+(b+d)i, \quad z_1-z_2=(a-c)+(b-d)i \tag{11}$$

$$(a+bi)(c+di)=(ac-bd)+(ad+bc)i, \quad \frac{z_1}{z_2}=\frac{z_1\overline{z_2}}{|z_2|^2}\ (z_2\neq 0) \tag{12}$$

$$\bar{z}=a-bi,\ z+\bar{z}=2\operatorname{Re}(z),\ z-\bar{z}=2i\operatorname{Im}(z),\ z\bar{z}=a^2+b^2=|z|^2 \tag{13}$$

$$|z|=\sqrt{a^2+b^2}, \quad |z_1z_2|=|z_1|\,|z_2|, \quad \left|\frac{z_1}{z_2}\right|=\frac{|z_1|}{|z_2|} \tag{14}$$

$$|z_1+z_2|\le |z_1|+|z_2| \text{ (uchburchak tengsizligi)} \tag{15}$$

$$\arg z=\theta+2k\pi,\ k\in\mathbb{Z}; \quad \operatorname{Arg}(z)\in(-\pi,\pi] \text{ bosh qiymat, } \tan\theta=\frac{b}{a} \text{ (chorakka e'tibor!)} \tag{16}$$

$$z=r(\cos\theta+i\sin\theta)=re^{i\theta}, \quad r=|z|,\ a=r\cos\theta,\ b=r\sin\theta \tag{17}$$

$$e^{i\theta}=\cos\theta+i\sin\theta \text{ (Euler)}, \quad e^{i\pi}+1=0 \tag{18}$$

$$z_1z_2=r_1r_2\,e^{i(\theta_1+\theta_2)}, \quad \frac{z_1}{z_2}=\frac{r_1}{r_2}\,e^{i(\theta_1-\theta_2)} \tag{19}$$

$$z^n=r^n\big(\cos(n\theta)+i\sin(n\theta)\big) \text{ (Muavr formulasi)} \tag{20}$$

$$z_k=r^{1/n}\left(\cos\frac{\theta+2k\pi}{n}+i\sin\frac{\theta+2k\pi}{n}\right), \quad k=0,\dots,n-1 \tag{21}$$

$$z^n=1\Rightarrow z_k=e^{2\pi ik/n}, \quad k=0,\dots,n-1 \ (n\text{-darajali birlik ildizlari}) \tag{22}$$

$$z=\frac{-b\pm\sqrt{b^2-4ac}}{2a} \text{ (kvadrat formula, } \mathbb{C} \text{ ustida ham o'rinli)} \tag{23}$$

$z^n=w=re^{i\theta}$ tenglamasining ildizlari (21) bilan beriladi; birlikning n-darajali ildizlari birlik aylanada muntazam n-burchak hosil qiladi. Haqiqiy koeffitsientli ko'phadlarning nohaqiqiy ildizlari qo'shma juftliklarda uchraydi, va Algebra asosiy teoremasiga ko'ra n-darajali ko'phadning C da ko'paytmalar hisobida aynan n ta ildizi bor.

2.2 Matematika I: 1-hafta amaliy mashqi — savollar va to’liq yechimlar

1. O’nli kasrlarning har birini chekli, periodik yoki aperiodik sifatida tasniflang va mos keladigan sonni ratsional yoki irratsional ekanligini ko’rsating: (1) 0.375; (2) 0.272727 . . .; (3) 1.101001000100001 . . .; (4) 3.14159265 . . .; (5) 0.123123123 . . .

[Manba: 1-hafta, amaliy mashq, masala 1]

Yechim. (1) 0.375 chekli, demak ratsional: $0.375 = \frac{375}{1000} = \frac{3}{8}$. (2) 27 bloki cheksiz takrorlanadi, demak ratsional; toza periodik blok uchun ab $= \frac{ab}{99}$, shuning uchun $0.27 = 27_{99} = 3_{11}$. (3) Ketma-ket birliklar orasidagi oraliqlar o’sib boradi (1, 2, 3, . . . nollar), shuning uchun kasr hech qachon periodik bo’lmaydi yoki to’lmaydi: son irratsional. (4) Ko’rsatilgan raqamlarning chekli soni nuqtalar bilan belgilangan sonni aniqlamaydi; agar $\pi = 3.14159265\ .\ .\ .$ nazarda tutilgan bo’lsa, u cheksiz va aperiodik, demak irratsional. (5) 123 bloki takrorlanadi, demak ratsional: $0.123 = \frac{123}{999} = \frac{41}{333}$. Prinsip: chekli yoki chekli muddatdan keyin periodik bo’lgan o’nli kasrlar aynan ratsional sonlardir.

2. Har bir to’plamni interval yozuvida yozing va sonli o’qda tasvirlang: (1) $\{x \in R : -3 \le x < 4\}$; (2) $\{x \in R : x \neq 3\}$.

[Manba: 1-hafta, amaliy mashq, masala 2]

Yechim. (1) -3 uchi kiritilgan (kvadrat qavs), 4 chiqarilgan (aylanma): $[-3, 4)$; o’qda -3 da to’liq nuqta, 4 da ochiq nuqta, ular orasidagi kesma belgilangan. (2) Barcha haqiqiy sonlar, 3 dan tashqari: $(-\infty, 3) \cup (3, \infty)$; o’qda butun o’q 3 nuqtasidan ochiq holda belgilangan. Cheksizlik hech qachon kiritilmaydi, shuning uchun har doim aylanma qavs ishlatiladi.

3. Har bir intervalni tengsizlik ko’rinishida yozing: a. $[-2, 5)$; b. $(-\infty, 4]$; c. $(1, \infty)$.

[Manba: 1-hafta, amaliy mashq, masala 3]

Yechim. a. $-2 \le x < 5$ (chap uchi kiritilgan, o’ng uchi chiqarilgan). b. $x \le 4$ (chap tomonga cheksiz, o’ng uchi kiritilgan). c. $x > 1$ (qat’iy katta, chunki aylanma qavs 1 ni chiqaradi). $\surd$

4. Ratsional yoki irratsional sifatida tasniflang: a. $\sqrt{3}$; b. $\sqrt{25}$; c. $\sqrt{50}$; i. $(1+\sqrt{5})(1-\sqrt{5})$.

[Manba: 1-hafta, amaliy mashg’ulot, topshiriq 4]

Yechim. a. $\sqrt{3}$ to’liq kvadrat emas, shuning uchun $\sqrt{3}$ irratsional. b. $\sqrt{25} = 5 \in Z \subseteq Q$, ratsional. c. $\sqrt{50} = 5\sqrt{2}$; $\sqrt{2}$ irratsional bo’lgani uchun, $5\sqrt{2}$ irratsional. i. Ochamiz: $(1+\sqrt{5})(1-\sqrt{5}) = 1 - 5 = -4$, ratsional. Konjugatlar ko’paytmasi har doim radikalni yo’q qiladi.

5. $z_1 = 3+$ 2i va $z_2 = 1-$ 4i berilgan; $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2 ni hisoblang, har bir javobni $a+$ bi ko’rinishida yozing.

[Manba: 1-hafta, amaliy mashq, 5-masala]

Yechim. Qo’shish va ayirish haqiqiy va mavhum qismlarga alohida ta’sir qiladi: $z_1 + z_2 = (3+1) + (2-4)i = 4-$ 2i; $z_1 - z_2 = (3-1) + (2+4)i = 2+$6i. Ko’paytma uchun $i^2 = -1va(a+bi)(c+di) = (ac-bd) + (ad+bc)i$ qoidasidan foydalanamiz: $z_1 z_2 = (3\cdot 1 - 2\cdot(-4)) + (3\cdot(-4) + 2\cdot 1)i = (3+8) + (-12+2)i = 11-$ 10i. Bo’linmada surat va maxrajni maxrajning kompleks konjugatiga ko’paytiramiz:

$$(z_1)/(z_2) = ((3+2i)(1+4i))/((1-4i)(1+4i))$$
$$(3+12i+2i+8i^2)/(1+16) = ((3-8)+(12+$$

$$2)i)/17 = (-5+14i)/17 = -\frac{5}{17} + (\frac{14}{17})i.$$

6. $3+$ **4i,** $-2+$ **5i, 6i, −7 sonlari uchun kompleks konjugatini va modulini toping; zz** $= |z|^2$ **tekshiring.** $\sqrt{a^2} + b^2$**.** $3+$ **4i:** $z = 3-$ **4i,** $|z| = 5$**,**

[Manba: 1-hafta, amaliy mashq, 6-masala]

Yechim. $z = a+$ bi bo'lsa, $z = a - biva|z| =$

$$\sqrt{-}2 + 5i : z = -2 - 5i, |z| = 4 + 25 =$$

va $(3-4i)(3+4i) = 9+16 = 25 = 5^2$. 29, va

$$6i : \bar{z} = -6i, |z| = 6, va(-6i)(6i) = -36i^2 = 36.$$

$(-2-5i)(-2+5i) = 4+25 = 29$. -7: $z = -7$ (haqiqiy sonlar o'zlari bilan konjugat bo'ladi), $|z| = 7, (-7)(-7) = 49$. Barcha to'rtta tekshiruv mos keladi.

1

7. $1 + nia+$ **bi ko'rinishida ifodalang.**

[Manba: 1-hafta, amaliy mashq, 7-masala]

Yechim. Maxrajning konjugatiga ko'paytiramiz:

$$1/(1+i) = (1-i)/((1+i)(1-i)) = (1-i)/2 = \frac{1}{2} - i/2$$

8. C **da yeching:** $x^2 + 9 = 0$**.**

[Manba: 1-hafta, amaliy mashq, 8-masala]

Yechim. Ko'chirib o'tganimizda, $x^2 = -9$ hosil bo'ladi, demak $x = \pm\sqrt{-}9 = \pm$3i. Ikkala yechim ham haqiqiy bo'lmagan kompleks sonlar; haqiqiy yechimlar yo'q, chunki R da kvadrat hech qachon manfiy bo'lmaydi.

9. $z = 2+$ **3i uchun kompleks tekislikda haqiqiy qism, mavhum qism, modul va kvadrantni aniqlang. 2 − 3i uchun nima o'zgaradi?**

[Manba: 1-hafta, amaliy mashq, 9-masala]

Yechim. (2, 3) nuqta birinchi kvadrantda joylashgan: $Re(z) = 2, Im(z) = 3, \sqrt{}$

$$|z| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

$|z| = 13$. $z = 2-$ 3i uchun (2, -3) nuqta haqiqiy o'q bo'yicha aks ettirilgan nuqta: haqiqiy qism va modul o'zgarmaydi, mavhum qism ishorasi o'zgaradi va nuqta to'rtinchi kvadrantga o'tadi.

10. $1+$ **ni trigonometrik shaklda** $z = r(\cos\theta + i\sin\theta)$ **va eksponensial shaklda** $z = re^{i\theta}$ **ko'rinishida yozing,** θ **radianda bo'lsin.**

[Manba: 1-hafta, amaliy mashq, 10-masala]

Yechim. Bu yerda $r = |1+i| = \sqrt{1}+1 = \sqrt{2}$, ikkala koordinatami ham musbat, $\sqrt{}$ shuning uchun $\theta = arctan(\frac{1}{1}) = \pi/4$. Shunday qilib, $1+i = 2\cos\frac{\pi}{4} + i\sin\pi = 2e^{i\pi/4}$. 4

11. Muavr formulasidan $[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin\theta)$ **foydalanib** $(\sqrt{3}+i)^4$ **ni hisoblang; algebraik tekshiring.**

[Manba: 1-hafta, amaliy mashq, 11-masala]

Yechim. Avval aylantiramiz: $r = |3+i| = \sqrt{3+1} = 2$, $\theta = arctan(1/\sqrt{3}) = \pi/6$. Muavr formulasiga ko'ra,

$$(\sqrt{3}+i)^4 = 2^4(\cos(4arctan(1/\sqrt{3})) + i\sin(4arctan(1/\sqrt{3})))$$
$$16(\cos(4\pi/6) + i\sin(4\pi/6)) =$$

$$16(\cos(2\pi/3) + i\sin(2\pi/3)) = 16(-\frac{1}{2} + \sqrt{\frac{3}{2}}i) = -8 + 8\sqrt{3}i.$$

Algebraik tekshirish:

$$(\sqrt{3}+i)^2 = 3 + 2\sqrt{3}i + i^2 = 2 + 2\sqrt{3}i;$$

yana kvadratga o'tkazsak,

$$(2+2\sqrt{3}i)^2 = 4 + 8\sqrt{3}i + 12i^2 = -8 + 8\sqrt{3}i.$$

Ikkala usul ham mos keladi.

12. $z^3 = -8$ **tenglamaning barcha yechimlarini toping va ularni kompleks tekislikda tasvirlang.**

[Manba: 1-hafta, amaliy mashg'ulot, 12-masala]

Yechim. Biz $-8 = 8\mathrm{e}^{i(\pi+2k\pi)}$ deb yozamiz. Kub ildizlari moduli $8^{1/3} = 2$ va argumentlari $(\pi+2k\pi)/3, k = 0,1,2$:

$$z_0 = 2e^{i\pi/3} = 1+\sqrt{3}i, z_1 = 2e^{i\pi} = -2,$$
$$z_2 = 2e^{i5\pi/3} = 1-\sqrt{3}i.$$

Geometrik jihatdan bu barchasi bir xil masofada joylashgan uchta nuqta, markazi koordinata boshida (radiusi 2) bo'lgan aylana bo'ylab (burchaklari $60^\circ, 180^\circ, 300^\circ$); $z_1 = -2-$ bu yagona haqiqiy yechim ekanligini ta'kidlaymiz, bu esa $(-2)^3 = -8ga$ mos keladi.

13. Har bir son uchun N, Z, Q, R, C **to'plamlaridan unga tegishli bo'lgan eng kichigini ko'rsating:** $\sqrt{0}$, -7, $\frac{5}{8}$, 3.25, 16, 7, $\sqrt{-9}$, π, π^2, i, $4-$ **3i.**

[Manba: 1-hafta, $ISA, 2$-topshiriq]

Yechim. $0 \in N$ (kurs kelishuviga ko'ra) $\Rightarrow$ eng kichik to'plam N. $-7 \in Z$. $\frac{5}{8} va 3.25 = \frac{13}{4} Q$ da joylashgan (chekli o'nli kasrlar). $\sqrt{16} = 4 \in N$. $\sqrt{7}$ irratsional (7 — to'liq kvadrat emas) $\Rightarrow R$. $-\sqrt{9} = -3 \in Z$. $\pi va \pi^2$ irratsional $\Rightarrow R$ (kvadratga ko'tarish irratsionalni ratsional qilmaydi). $iva 4-$ 3i haqiqiy emas $\Rightarrow$ faqat C.

14. To'g'ri yoki noto'g'ri? Har bir noto'g'ri da'vuni tuzating: $(1) N \subseteq Z$**;** $(2) Z \subseteq Q$**;** $(3) R \subseteq Q$**;** $(4) Q \subseteq R$**;** $(5) R \subseteq C$**;** $(6) har$ **bir butun son natural;** $(7) har$ **bir**

ratsional son haqiqiy; $(8) har$ **bir kompleks son haqiqiy;** $(9) har$ **bir haqiqiy sonni kompleks sifatida yozish mumkin;** $(10) son$ **bir vaqtda ratsional va haqiqiy bo'la olmaydi.** $\surd$

[Manba: 1-hafta, $ISA, 3$-topshiriq]

Yechim. (1) To'g'ri. (2) To'g'ri. (3) Noto'g'ri — masalan, $2 \in R$, lekin $\sqrt{2}/ \in Q$; to'g'ri inkluziya $Q \subseteq R$. (4) To'g'ri. (5) To'g'ri. (6) Noto'g'ri — manfiy butun sonlar, masalan -3, natural emas. (7) To'g'ri. (8) Noto'g'ri — masalan, i kompleks, lekin haqiqiy emas. (9) To'g'ri — $a = a+$ 0i. (10) Noto'g'ri — har bir ratsional son haqiqiy (da'v (4)), shuning uchun ratsional sonlar ikkalasiga ham tegishli.

15. $N \subset Z \subset Q \subset R \subset C$ **to'plamlarining tushuntirish diagrammasini chizing va** $\sqrt{5}, -2, \frac{2}{3}, 2, \pi, -$**4i,** $3+$ **2i ni to'g'ri joylariga joylashtiring.**

[Manba: 1-hafta, $ISA, 4$-topshiriq]

Yechim. Beshta tushuntirish ovalini chizamiz, N (ichki)$dan\sqrt{C}$(tashqi) gacha belgilar bilan. Joylashtirish: $5 \in N$; $-2 \in Z \;\; N$; $\frac{2}{3} \in Q \;\; Z$; $2va\pi \in R \;\; Q$; $-$4i va $3+$ 2i $\in C \;\; R$. Diagramma har yangi to'plam faqat sonlar qo'shilishini, hech narsani olib tashlamasligini ko'rsatadi. $\surd$

16. O'sish tartibida joylashtiring: $-5, -2, -\frac{3}{2}, 0, 2, \frac{3}{2}, \frac{\pi}{2}$**; noaniq taqqoslashlarni asoslang.** $\surd$

$$(\frac{2}{3})^2 = \frac{4}{9} < (\pi/2)^2$$

[Manba: 1-hafta, $ISA, 5$-topshiriq]

Yechim. Kvadratlar musbat qiymatlar tartibini beradi: $\sqrt{\frac{3}{4}} = (\sqrt{3})/2 \approx 0.866$; $\pi/2 \approx 1.5708 va \frac{3}{2} = 1.5$ ekanligidan, $\frac{3}{2} < \pi/2$. Manfiy sonlar uchun tartib teskari: $-\sqrt{5} \approx -2.236 < -2 < -\frac{3}{2}$. Yakunda: $-\sqrt{5} < -2 < -\frac{3}{2} < (\sqrt{3})/2 < \frac{3}{2} < \pi/2$.

$$-\sqrt{5} < -2 < -\frac{3}{2} < 0 < 2 < 3 < \pi/2$$

17. Har bir to'plamni interval yozuvida ifodalang: $(1) -3 \leq x < 4$**;** $(2) x > 2$**;** $(3) x \leq -1$**;** $(4) -2 < x \leq 5$**;** $(5) x \neq 3$**; keyin har birini chiziqda tasvirlang.**

[Manba: 1-hafta, $ISA, 6$-topshiriq]

Yechim. (1) $[-3, 4)$. $(2)(2, \infty)$. $(3)(-\infty, -1].(4)(-2, 5].(5)(-\infty, 3) \cup (3, \infty)$. Chiziqda: cheki kiritilgan joyda $(\leq, \geq)$ nuqta to'ldirilgan, kiritilmaganida $(<, >)$ ochiq, cheksizliklarda qavs va (5) da ikkita nur.

18. Har bir intervalni tengsizlik ko'rinishida yozing: a**.** $[-2, 5)$**;** b**.** $(-\infty, 4]; c.(1, \infty); d.(-3, 0] \cup [2, 6); e.R\ [-1, 1]$.

[Manba: 1-hafta, $ISA, 7$-topshiriq]

Yechim. a. $-2 \leq x < 5$. b. $x \leq 4$. c. $x > 1$. d. $-3 < x \leq 0$ yoki $2 \leq x < 6$. e. $x < -1$ yoki $x > 1$; $|x| > 1$ ga ekvivalent. $\surd$

19. $d(a, b) = |a - b|$ **bilan:** $d(3, 8), d(-2, 5), d(-4, -9), d(2, 8)$ **ni hisoblang;** $a = -3$, $b = 2$, $c = 7dad(a, c) \leq d(a, b) + d(b, c)$ **tekshiring.**

$$d(2, 8) = |2 - 82| = 2$$

[Manba: 1-hafta, $ISA, 8$-topshiriq]

Yechim. $d(3,8)=|3-8|=5$; $d(-2,5)=7$; $d(-4,-9)=5$; $d(2,8)=|2-22|=2$. Uchburchak tengsizligini tekshirish: $d(-3,\,7)=10$, $d(-3,\,2)+d(2,\,7)=5+5=10$, va $10\le 10$ (tenglik, chunki nuqtalar bir chiziqda ketma-ket joylashgan). $|a-b|$ formulasi (28) dan olingan bir o'lchamli norma holati.

20. Har bir sonni ratsional yoki irratsional ekanligini aniqlang; asoslang: a**.** $\sqrt{49}$**;** b**.** $\sqrt{18}$**;** c**.** $\sqrt{3}+\sqrt{2}$**;** d**.** $\sqrt{2}\cdot\sqrt{5}$**;** e**.** $\sqrt{2}+\sqrt{8}$**;** f**.** $(\sqrt{3})^2$**;** g**.** $(1+\sqrt{2})(1-\sqrt{2})$**;** h**.** $\sqrt{12}$**.**
[Manba: 1-hafta, ISA, topshiriq 9]

Yechim. a. $\sqrt{49}=7$ — ratsional. b. $\sqrt{18}=3\sqrt{2}$ — irratsional. c. $\sqrt{3}+\sqrt{2}$ — irratsional. d. $\sqrt{2}\cdot\sqrt{5}=\sqrt{10}$ — irratsional. e. $\sqrt{2}+\sqrt{8}=\sqrt{2}+2\sqrt{2}=3\sqrt{2}$ — irratsional. f. $(\sqrt{3})^2=3$ — ratsional. g. $(1+\sqrt{2})(1-\sqrt{2})=1-2=-1$ — ratsional (konjugat ko'paytma). h. $\sqrt{12}=2\sqrt{3}$ — irratsional. Amaliy qoida: soddalashtirish tabiatni ochib beradi; $(a+\sqrt{b})(a-\sqrt{b})$ ko'rinishidagi konjugat ko'paytmalar har doim ratsional.

21. $z_1=3+$ **2i,** $z_2=1-$ **4i berilgan;** $z_1+z_2,\ z_1-z_2,\ z_1z_2,\ z_1/z_2$ **hisoblang.**
[Manba: 1-hafta, ISA, topshiriq 10]

Yechim. Yuqoridagi amaliy mashq 5 bilan bir xil: $z_1+z_2=4-$ 2i, $z_1-z_2=2+$ 6i, $z_1z_2=11-$ 10i, va $z^1=-175+1417i$.

$$z_2=1-4i$$

22. $i^2,\ i^3,\ i^4,\ i^5,\ i^{18},\ i^{27},\ i^{100}$ **ni hisoblang; takrorlanadigan naqshni va 4 ga bo'lishga asoslangan usulni tavsiflang.**
[Manba: 1-hafta, ISA, topshiriq 11]

Yechim. $i^2=-1$; $i^3=i^2i=-i$; $i^4=(i^2)^2=1$; $i^5=i$. Darajalar 4 davr bilan siklik: $1,i,-1,-i,\ldots$ Katta daraja uchun 4 ga bo'lib qoldiq r ni topamiz: $i^n=i^r$. Unda $18=4\cdot4+2\Rightarrow i^{18}=i^2=-1$; $27=4\cdot6+3\Rightarrow i^{27}=i^3=-i$; $100=4\cdot25\Rightarrow i^{100}=i^0=1$.

23. $3+$ **4i,** $-2+$ **5i, 6i, −7 uchun: konjugatini va modulini toping,** $z\bar{z}=|z|^2$ **tekshiring.**
[Manba: 1-hafta, ISA, topshiriq 12]

Yechim. Yuqoridagi amaliy mashq 6 dagi kabi hisoblar: $\surd$ konjugatlar 3 − 4i, −2 − 5i, −6i, −7; modullar 5, 29, 6, 7; har bir holatda $z\bar{z}=a^2+b^2=|z|^2$ aniq bajariladi.

24. $a+$ **bi ko'rinishida ifodalang:** a**.** $1/(1+i)$**;** b**.** $3+2/(2-i)$**;** c**.** $1+3/(4-i)$**;** d**.** $5/(2-i)$**.**
[Manba: 1-hafta, ISA, topshiriq 13]

Yechim. Har bir holatda surat va maxrajni maxrajning konjugatiga ko'paytiramiz. a.

$$(1+i)/(1-i)=(1+i)^2/((1-i)(1+i))=(1+2i+i^2)/(1-i^2)$$
$$(1+2i-1)/(1-(-1))=(2i)/2=i$$

$=i$. b.

$$(3+2i)/(2-i)=(3+2i)(2+i)/((2-i)(2+i))=(6+3i+4i+2i^2)/(4-i^2)$$
$$(6+7i-2)/(4-(-1))=(4+7i)/5$$

$= \frac{4}{5} + \frac{7}{5}i$. *c.*

$$(1+3i)/(4-i) = (1+3i)(4+i)/((4-i)(4+i)) = (4+i+12i+3i^2)/(16-i^2)$$
$$(4+13i-3)/(16-(-1)) = (1+13i)/17$$

$= \frac{1}{17} + \frac{13}{17}i$. *d.*

$$5i/(2-i) = 5i(2+i)/((2-i)(2+i)) = (10i+5i^2)/(4-i^2)$$
$$(10i-5)/(4-(-1)) = (-5+10i)/5$$

= -1+ 2i.

25. C **ustida yeching, qaysi yechilar haqiqiy ekanligini ko'rsating:** *a.* $x^2+9=0$**;** *b.* x^2- **4x** $+13=0$**;** *c.* **2x**$^2+$ **4x** $+10=0$**;** *d.* $x^4-16=0$**.**

[Manba: 1-hafta, ISA, topshiriq 14]

Yechim. *a.* $x = \pm$3i (haqiqiy emas). *b.* Diskriminant $\Delta = (-4)^2 - 4\cdot 13 = 16-52 = -36$, $\sqrt{\Delta} =$ 6i; $x = (4 \pm 6i)/2 = 2\pm$ 3i (haqiqiy emas). *c.* 2 ga bo'lamiz:

$$x = (-4 \pm \sqrt{4^2 - 4\cdot 2\cdot 10})/(2\cdot 2) = (-4 \pm \sqrt{-72})/4 = (-4 \pm 6i\sqrt{2})/4 = -1 \pm (3i\sqrt{2})/2$$

x^2+ 2x $+5=0$: $\Delta = 4-20 = -16$, $\sqrt{\Delta} =$ 4i; $x = (-2\pm 4i)/2 = -1\pm$ 2i (haqiqiy emas).

d. $x^4 = 16$ haqiqiy yechilarni beradi $x = \pm 2$; C ustida to'liq to'plam — 16 ning to'rtinchi ildizlari: 2, −2, 2i, −2i (barcha yechilar $(x^2-4)(x^2+4)=0$).

26. $2+$**3i,** $-1+$**4i, −3 − 2i,** $4-i$ **uchun: haqiqiy qismini, mavhum qismini, modulini, kvadrantini toping.** $\surd$

[Manba: 1-hafta, ISA, topshiriq 15]

Yechim. $2+$ 3i: (2, 3), modul $\sqrt{13}$, kvadrant I. $-1+$ 4i: (−1, 4), modul 17, $\surd$ kvadrant II. −3 − 2i: (−3, −2), modul $\sqrt{13}$, kvadrant III. $4-i$: (4, −1), modul 17, kvadrant IV. E'tibor bering, modullar juftliklarga bo'linadi, chunki konjugatsiya a^2+b^2 ni saqlaydi.

27. Trigonometrik va eksponensial shaklda (θ radianda yozing): $(1)\sqrt{1}+i$**;** $(2)-1+i$**;** $(3)-3-i$**; (4) 2i; (5) −4.** $\surd$

[Manba: 1-hafta, ISA, 16-masala]

Yechim. $(1) r = 2, \theta = \pi/4$: 2e$^{i\pi/4}$. $(2) r = 2$, kvadrant II: $\theta = 3\pi/4$: 2e$^{i3\pi/4}$. $(3) r = \sqrt{3+1} = 2$, kvadrant III: asosiy burchak $arctan(1/\sqrt{3}) = \pi/6$, demak $\theta = \pi + \pi/6 = 7\pi/6$: 2e$^{i7\pi/6}$. $(4) r = 2, \theta = \pi/2$: 2e$^{i\pi/2}$. $(5) r = 4, \theta = \pi$: 4e$^{i\pi}$. Har bir burchak $\tan\theta = b/a$ dan kvadrantga qarab tuzatib olinadi, (16) da ogohlantirilgandek. $\surd$

28. Muavr formulasi yordamida hisoblang $(1)(1+i)^6$**;** $(2)(3+i)^4$**;** $(3)(-1+3i)^5$**; bittasini algebr tekshiring.** $\surd$

$$(1+i)^6 = (\sqrt{2})^6 e^{i6\pi/4} = 8e^{i3\pi/2} = 8(0-i) = -8i.$$

[Manba: 1-hafta, ISA, topshiriq 17]

Yechim. $(1) 1+i = \sqrt{2}e^{i\pi/4}$. U holda $(1+i)^6 = (\sqrt{2})^6 e^{i6\pi/4} = 8e^{i3\pi/2} = 8(0-i) = -$8i. (2) Amaliy mashq 11 dan: $(3+i)^4 =$ -8 $+ 8\sqrt{3}i$. $(3) z = -1+$ 3i: modul $|z| = \sqrt{1+3} = 2$; II chorak, asosiy burchak $\pi/3$, demak $\theta = 2\pi/3$: $z =$ 2e$^{i2\pi/3}$. U holda $z^5 = 2^5 e^{i10\pi/3} =$

$32\mathrm{e}^{i(10\pi/3-2\pi)} = 32\mathrm{e}^{i4\pi/3} = 32(-\frac{1}{2} - i\sqrt{3}/2) = -16 - 16\sqrt{3}i$. Tekshirish (1) algebra: $(1+i)^2 = 2\mathrm{i}$, $(1+i)^4 = (2i)^2 = -4$, $(1+i)^6 = -4\cdot 2\mathrm{i} = -8\mathrm{i}$. Mos keladi.

29. Barcha yechimlarni toping va ularni kompleks tekislikda tasviring: a**.** $z^3 = 1$**;** b**.** $z^4 = 1$**;** c**.** $z^3 = -8$**. Tushuntiring,** n**-darajali ildizlari nima uchun aylana bo‘yiga teng masofada joylashgan.** $\surd$

[Manba: 1-hafta, ISA, topshiriq 18]

Yechim. a. (22) ga ko‘ra: $z_k = e^{2\pi ik/3}, k = 0,1,2$: 1, $-\frac{1}{2} + \frac{3}{2}i$, $-\frac{1}{2} - \frac{3}{2}i$. b. $z_k = e^{2\pi ik/4}$: $1, i, -1, -i$. c. Amaliy mashq 12 dan: $\sqrt{1}+$ 3i, -2, $1 -$ 3i. Barcha holatlarda ildizlarning bir xil modulga ega $r^{1/n}$ (demak, bir aylanada joylashgan), ularning argumentlari aniq $\frac{2\pi}{n}$ farq qiladi (to‘liq aylanish $2\pi n$ teng qismga bo‘lingan) — shu sababli ular muntazam n-burchakli uchburchakning uchlari hosil qiladi.

30. Elektr zaryadi miqdori $e = 0.000000000000000000001602176634Kl$**. Uni ilmiy va muhandislik yozuvida yozing.**

[Manba: 1-hafta, ISA, topshiriq 19]

Yechim. 18 xonali hisoblaymiz, birinchi nol bo‘lmagan raqam vergul oldiga chiqsin: $e = 1.602176634 \times 10^{-19} Kl$ (ilmiy). Muhandislik yozuvida daraja 3 ga karrali bo‘lishi kerak, shuning uchun bitta xonaga siljitamiz: $e = 160.2176634 \times 10^{-21} Kl =$ 160.2176634 zKl (zeptokulon); amaliyotda $160.2176634 \times 10^{-21} Kl$ deb yoziladi, chunki 10^{-24} dan pastdagi SI old qo‘shimchalari faqat zepto (10^{-21}) mavjud.

31. SI **old qo‘shimchalari jadvalini to‘ldiring:** 10^{12}, 10^{6}, 10^{3}, 10^{-6}, 10^{-9}, 10^{-12} **darajalariga mos old qo‘shimcha/belgilar.**

[Manba: 1-hafta, ISA, topshiriq 20]

Yechim. 10^{12} tera (T); 10^{6} mega (M); 10^{3} kilo (k); 10^{-6} mikro (µ); 10^{-9} nano (n); 10^{-12} piko (p). Giga (G), milli (m), femto (f), atto (a) ma’lumotlari bilan birga bu jadval to‘liq SI zanjirini tashkil qiladi.

32. 1-hafta konspekti, 2-qismi, auditoriya masalalari: (1) 0.00000824 ni ilmiy va muhandislik yozuvida yozing; $(2) 7.35 \times 10^8 Gz$ **ni muhandislik yozuvida yozing;** $(3) 0.0047 Fni$ **old qo‘shimcha bilan yozing.**

[Manba: 1-hafta, konspekt, 2-qism]

Yechim. $(1) 0.00000824 = 8.24 \times 10^{-6}$ (ilmiy); -6 3 ga karrali, shuning uchun bu allaqachon muhandislik yozuvi, masalan 8.24 µA, agar birlik amper bo‘lsa. $(2) 10^8$3 ga karrali emas; $7.35 \times 10^8 = 735 \times 10^6 = 735$ MGz deb yozamiz. $(3) 0.0047 = 4.7 \times 10^{-3} =$ 4.7 mF.

33. 1-hafta konspekti, 3-qismi, auditoriya masalalari: $(1) N \subset Z \subset Q \subset R \subset C$ **kiritilgan to’plamlar diagrammasini chizing va** 2026, -9, $\frac{2}{3}$, 10, π, $-$**5i,** $-2+$ **2i ni joylashtiring; (2) ratsional yoki irratsional:** a**.** $\sqrt{49}, b$**.** $\sqrt{18}, c$**.** $\sqrt{3} + \sqrt{2}, d$**.** $\sqrt{2} \cdot \sqrt{5}, e$**.** $\sqrt{2} + \sqrt{8}, f$**.** $(\sqrt{5})^2$**.**

[Manba: 1-hafta, konspekt, 3-qism]

Yechim. $(1) 2026 \in N$; $-9 \in Z \quad N$; $\frac{2}{3} \in Q \quad Z$; $10 va \pi \in R \quad Q$; $-$5i va $-2+$ 2i $\in C \quad R$. $(2) a$. $\sqrt{49} = 7$ — ratsional. b. $\sqrt{18} = 3\sqrt{2}$ — irratsional. c. $\sqrt{3} + \sqrt{2}$ — irratsional. d. $\sqrt{2} \cdot \sqrt{5} = \sqrt{10}$ — irratsional. e. $\sqrt{2} + \sqrt{8} = 3\sqrt{2}$ — irratsional. f. $(\sqrt{5})^2 = 5$ — ratsional — 9-masaladagi bir xil mantiq, bu mashq boshqa sonlar bilan takrorlanadi.

34. 1-hafta konspekti, 3-qismi, auditoriya masalalari: $z_1 = 2+$ **3i,** $z_2 = 1-$ **5i berilgan;** $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1 / z_2 **ni hisoblang.**

[Manba: 1-hafta, konspekt, 3-qism]

Yechim. $z_1+z_2=3-$ 2i; $z_1-z_2=1+$ 8i; ko‘paytma: $(2\cdot1-3\cdot(-5))+(2\cdot(-5)+3\cdot1)i=(2+15)+(-10+3)i=17-$ 7i; bo‘linma:

$$(z_1)/(z_2)=((2+3i)(1+5i))/((1-5i)(1+5i))$$
$$(2+10i+3i+15i^2)/(1-(5i)^2)=(2+13i-15)/(1+$$

$$25)=(-13+13i)/26=-\frac{1}{2}+\frac{1}{2}i.$$

35. Tushuntiring, nima uchun $1+i3=2\cos\frac{\pi}{3}+i\sin\pi=$ 2e$^{i\pi/3}$. 3

$$|1+i\sqrt{3}|=\sqrt{1^2+(\sqrt{3})^2}=\sqrt{1+3}=2.$$

[Manba: 1-hafta, konspekt, qism 3]

Yechim. Modul $|1+i$ Nuqta (1, 3) birinchi chorakda joylashgan, bunda $\tan\theta=3$, shundan $\theta=\pi/3$. $r=2,\theta=\pi/3ni(17)$ polar shakliga, keyin Eyler formulasi (18) ga qo’yib, ikkala yozilgan tenglamalarni hosil qilamiz.

3 2-hafta — Funksiyalar (Math I)

3.1 Math I: 2-hafta formulalari katalogi

3.1.1 Funksiyalar: ta’rif, sinflar, simmetriya, kompozitsiya, teskari funksiyalar

Funksiya sohasidagi har bir x ga aynan bitta natija $f(x)$ ni bog’laydi; vertikal to’g’ri testi bunga mos kelmaydigan bog’lanishlarni aniqlaydi. Tabiiy soha — formula mazmunli bo’ladigan haqiqiy kirishlarning eng katta to’plami; uchta standart to’siq: nolga bo’lish, manfiy sonlarning juft darajali ildizlari va nomanfiy bo’lmagan sonlarning logarifmi.

$$f:D\to\mathbb{R},\ x\mapsto f(x);\quad D=\text{soha},\ \text{range}\,f=\{f(x):x\in D\} \tag{31}$$

$$\frac{1}{g(x)}:g(x)\neq0,\quad \sqrt{g(x)}:g(x)\geq0,\quad \log_b g(x):g(x)>0 \tag{32}$$

Juftlik va toqlik mos ravishda y-o’qqa va bosh nuqtaga nisbatan simmetriyani aks ettiradi; kompozitsiya funksiyalarni zanjirlaydi; faqat bir-biriga bir xil bog’lanmagan (gorizontal to’g’ri testi) funksiyalarga teskari funksiya mavjud bo’lib, uni $y=f(x)nix$ ga nisbatan yechib topiladi.

$$\text{juft: } f(-x)=f(x),\quad \text{toq: } f(-x)=-f(x) \tag{33}$$

$$(f\circ g)(x)=f\big(g(x)\big),\quad (f\circ g)\neq(g\circ f)\text{ umuman olganda} \tag{34}$$

$$f\text{ inyektiv}\Leftrightarrow f(x_1)=f(x_2)\Rightarrow x_1=x_2 \tag{35}$$

$$f^{-1}\big(f(x)\big) = x \text{ va } f\big(f^{-1}(y)\big) = y \tag{36}$$

$$f^{-1} \text{ grafigi} = f \text{ grafigining } y = x \text{ dagi aksi} \tag{37}$$

§1.2 (OpenStax) funksiya sinflari: chiziqli, ko'phad, ratsional, ildiz (algebraik), eksponensial/logarifmik va trigonometrik (transsendent); har bir ko'phad hamma joyda silliq, va har bir ratsional funksiya maxrajnining noldan farqli bo'lgan joylarida aniqlangan.

3.1.2 Eksponensial va logarifmik funksiyalar (§1.5, giperbolik funksiyalar chiqarilgan)

Eksponensial va logarifm bir-birining teskarisi; daraja va logarifm qonunlari ko'paytirishni qo'shishga aylantiradi va kursning har bir o'sish/pasayish modelida ishlatiladi.

$$b^x > 0, \quad b^{x+y} = b^x b^y, \quad (b^x)^y = b^{xy}, \quad b^{-x} = \frac{1}{b^x} \tag{38}$$

$$\log_b x = y \Leftrightarrow b^y = x, \quad x > 0,\ b > 0,\ b \neq 1 \tag{39}$$

$$\log_b(xy) = \log_b x + \log_b y, \quad \log_b \frac{x}{y} = \log_b x - \log_b y, \quad \log_b x^n = n \log_b x \tag{40}$$

$$\log_b x = \frac{\ln x}{\ln b} \text{ (asosni almashtirish)}, \quad b^x = e^{x \ln b} \tag{41}$$

$$e^{\ln x} = x, \quad \ln 1 = 0, \quad \ln e = 1 \tag{42}$$

O'sish va pasayish modellari $A(t) = Aoa^tniAoe^{kt}$ bilan o'sish tezligi konstantasi orqali taqqoslaydi; ikki barobar o'sish vaqti — me'yoriy kattalik.

$$A(t) = A_0 a^t = A_0 e^{kt}, \quad k = \ln a, \quad a > 1:\ \text{o'sish};\ 0 < a < 1:\ \text{pasayish} \tag{43}$$

$$N(t) = N_0 2^{t/T_d} = N_0 e^{kt}, \quad k = \frac{\ln 2}{T_d}, \quad T_d = \frac{\ln 2}{k} \tag{44}$$

Daraja va radianlar $\pi rad = 180°$ orqali o'zgartiriladi:

$$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$$

3.2 Math I: 2-hafta amaliy mashg'uloti (funksiyalar) — savollar va to'liq yechimlar

1. Har bir munosabat $ynix$ funksiyasi sifatida belgilaydimi: a. $y =$ 3x −

$$x^2 + y^2 = 9.$$

2; b. U yerga tegishli bo'lsa, y bo'yicha yeching va vertikal to'g'ri chiziq testini qo'llang.

[Manba: 2-hafta, amaliy mashg'ulot, 1-masala]

Yechim. $a.$ $y = 3\text{x} - 2harbirx$ uchun ayni bir y ni beradi: funksiya (to'g'ri chiziq). $\sqrt{b}.$ Yechish orqali $y = \pm 9 - x^2$ hosil bo'ladi: ko'p x (masalan $x = 0$) ikkita $y = \pm 3$ qiymatini beradi, shuning uchun vertikal to'g'ri chiziq aylana bilan ikki marta kesishadi; funksiya emas.

2. $f(x) = \mathbf{2x}^2 - \mathbf{3x} + 1$ **deb belgilaymiz. Hisoblang va soddalashtiring:** $f(0), f(-2),$ $f(a),\ f(a+h)vaf(a+h) - f(a),\ h \neq 0$**.**

[Manba: 2-hafta, amaliy mashg'ulot, 2-masala]

Yechim. $f(0) = 1;\ f(-2) = 2(4) + 6 + 1 = 15;\ f(a) = 2\text{a}^2 - 3\text{a} + 1;\ f(a+h) = 2(a+h)^2 - 3(a+h) + 1 = 2\text{a}^2 + 4\text{ah} + 2\text{h}^2 - 3\text{a} - 3\text{h} + 1.$ Farq nisati:

$$(f(a+h) - f(a))/h = (4ah + 2h2 - 3h)/h = 4a + 2h - 3.$$

Bu ifoda 3-4 haftalardagi hosilani ($h \to 0$ bo'lganda $4\text{a} - 3$ ga intiladi) oldindan aytib beradi.

$$:_{x2+9}; b.g(x) = \sqrt{5} - 2x.$$

3. Tabiiy aniqlanish sohasini toping: $a.$ $f(x) =^{x-1}$ **Interval yozuvida yozing. Maxraj** $x^2 + 9 \geq 9 > 0$ **hech qachon nolga teng emas,**

[Manba: 2-hafta, amaliy mashg'ulot, 3-masala]

Yechim. $a.$ shuning uchun soha — barcha $R = (-\infty, \infty).$ $b.$ $5 - 2\text{x} \geq 0 \Longleftrightarrow x \leq \frac{5}{2}$ kerak: soha $(-\infty, \frac{5}{2}]$.

4. $f(x) = \mathbf{2x} - 3,\ g(x) = x^2 + 1$ **deb belgilaymiz.** $(f \circ g)(x)va(g \circ f)(x)$ **ni toping; ular tengmi?**

[Manba: 2-hafta, amaliy mashg'ulot, 4-masala]

Yechim. $(f \circ g)(x) = f(g(x)) = 2(x^2+1) - 3 = 2\text{x}^2 - 1;\ (g \circ f)(x) = g(2x-3) = (2x-3)^2 + 1 = 4\text{x}^2 - 12\text{x} + 10.$ Ular farqli (masalan, $x = 0$ da: $-1va10$), bu tasdiqlaydiki: kompozitsiya umuman olganda kommutativ emas — avval kvadratga ko'tarib, keyin siljitish bilan avval siljitib, keyin kvadratga ko'tarish bir xil emas.

5. Har bir funksiyani juft, toq yoki umumiy ko'rinishda ekanligini aniqlang: $|x| + 1va\frac{x}{x2+1}$**; algebraik va grafik tekshiring.**

[Manba: 2-hafta, amaliy mashg'ulot, 5-masala]

Yechim. $f(x) = |x| + 1$: $f(-x) = |-x| + 1 = |x| + 1 = f(x)$ — juft; grafik y o'qi bo'yicha aks ettirilma simmetriyaga ega. $g(x) = \frac{x}{x2+1}$: $g(-x) = \frac{-x}{x2+1} = -g(x)$ — toq; grafik koordinata boshi atrofida 180° burilganda simmetrik.

6. Har bir funksiyani chiziqli, ko'phadli, ratsional, algebraik yoki transsendent (funksiya bir nechta toifalarga tegishli bo'lishi mumkin) sifatida tasniflang: $_{x2+4}$ **va**

[Manba: 2-hafta, amaliy mashg'ulot, 6-masala]

Yechim.$^{x+1}$ — ko'phadlar nisbati, demak ratsional $(va$

$$\sqrt{x^2 + 4}$$

$x^2 + 1$ — ko'phadning ildizi, demak algebraik algebraik). (ratsional emas, chunki ko'phadlar nisbati emas).

7. Sanoat kamerasidagi harorat $t = 0da20^\circ Cdant = 15da80^\circ C$ **gacha chiziqli o‘sadi.** $(1)T(t)$ **ni toping; (2) qiyalik va siljishni tushuntiring;** $(3)T(8)$ **ni toping; (4) kamera qachon** $60^\circ C$ **ga yetadi? (5) fizik jihatdan mantiqiy aniqlanish sohasini ko‘rsating.**

[Manba: 2-hafta, amaliy mashg‘ulot, 7-masala]

Yechim. (1) Qiyalik $m = (80-20)/(15-0) = 4$ °C/daq, demak $T(t) = 20+$ 4t. (2) Qiyalik — isitish tezligi: daqiga $+4$ gradus; siljish 20 — boshlang’ich harorat. $(3)T(8) = 20 + 32 = 52$ °C. $(4)20+$ 4t $= 60dat = 10$ daqiqa.

8. $p(x) = x^3-$ **4x uchun aniqlang: aniqlanish sohasi; o‘qlar bilan kesishish; simmetriya; chekka xatti-harakatlar; noaniqlik nuqtalari.**

[Manba: 2-hafta, amaliy mashg‘ulot, 8-masala]

Yechim. Aniqlanish sohasi R (ko‘phad). Kesishish: $p(0) = 0$; x^3- 4x $= x(x-2)(x+2) = 0x = -2,\ 0,\ 2$ beradi — barchasi koordinata boshidan o‘tuvchi x o‘qi bilan. Simmetriya: $p(-x) = -x^3+$4x $= -p(x)$ — toq. Chekka xatti-harakatlar: yuqori darajali had x^3 musbat koeffitsientga ega, shuning uchun $p(x) \to +\infty x \to +\infty vap(x) \to -\infty x \to -\infty$ da. Ko‘phad hamma joyda aniqlangan: noaniqlik nuqtalari yo‘q.

9. Elektron ta’minot yetkazib beruvchisi $0 \le E \le 200$ **uchun** $C(E) = 0.10EvaE > 200$ **uchun** $C(E) = 20 + 0.15(E - 200)$**, kVt·**$ch$ **da.** $(1)C(150),\ C(200),\ C(350)$ **ni toping; (2) tasvirlang; (3) model** $E = 200$ **da uzluksizmi?** $(4)har$ **bir qiyalikning mohiyati; (5) bir cheklov.**

[Manba: 2-hafta, amaliy mashg‘ulot, 9-masala]

Yechim. $(1)C(150) = 15$; $C(200) = 20$; $C(350) = 20+0.15\cdot 150 = 42.5$ (barchasi \$ da). (2) Ikkita chiziqli qism: 200 gacha qiyalik 0.10, keyin 0.15 — (200, 20) da burchak hosil qiladigan qisml-chiziqli grafik. $(3)Ha$: ikkala formula ham $E = 200da20$ beradi, qismlar saklamasdan tutashadi. (4) Qiyaliklar — chekiya narxlari: birinchi 200 kVt·ch uchun 0.10\$/kVt·ch, undan ortig‘i uchun 0.15\$/kVt·ch. (5) Cheklov: narxning aynan 200 kVt·ch da saklanishi iste’molchini chegaradan biroz pastroqda to‘xtashga undaydi; haqiqiy tariflar asta-sekin yoki bloklar bilan o‘zgaradi.

10. $f(x) =$ **4x − 7 ga teskari funksiyani toping va aniqlanish sohalarini ko‘rsating;** $f(f^{-1}(x)) = x$ **ekanligini tekshiring.**

[Manba: 2-hafta, amaliy mashg‘ulot, 10-masala]

Yechim. $y =$ 4x $-7 \Rightarrow x = (y + 7)/4$, shuning uchun $f^{-1}(x) = (x + 7)/4$, aniqlanish sohasi R va qiymatlar to’plami R. Tekshirish: $f(f^{-1}(x)) = 4\cdot(x+7)/4-7 = x+7-7 = x$. Ikkalasi ham to’g’ri chiziqlar (grafiklar $y = x$ ga nisbatan simmetrik).

11. Har bir model uchun o’sish/yemirilishni aniqlang va boshlang’ich qiymatni hamda o’sish/yemirilish koeffitsiyentini ko’rsating: $A(t) = 500(1.08)^t$, $B(t) = 1200(0.85)^t$, $C(t) =$ **40e**$^{0.03t}$, $D(t) =$ **200e**$^{-0.4t}$**.**

[Manba: 2-hafta, amaliy mashg’ulot, topshiriq 11]

Yechim. A: o’sish, boshlang’ich 500, koeffitsient 1.08(+%8 davr uchun). B: yemirilish, boshlang’ich 1200, koeffitsient 0.85 (−15%). C: o’sish, boshlang’ich 40, uzluksiz stavka $k = 0.03$. D: yemirilish, boshlang’ich 200, uzluksiz stavka $k = -0.4$. e^{kt} dagi k belgisi yoki a^t dagi $a > 1$ bo’lishi yo’nalishni belgilaydi.

12. $2^{x+1} = 16$ **tenglamani yeching.**

[Manba: 2-hafta, amaliy mashg’ulot, topshiriq 12]

Yechim. $16 = 2^4$, shuning uchun $x + 1 = 4vax = 3$. (Teng ikkala tomondan $\log_2$ olish orqali ham yechish mumkin.)

13. Batareyaning zaryadi: $E(t) = E_0 e^{-0.025t}$, t **soatda. (1) 10 soatdan keyin qolgan energiya ulushi? (2) 50% qolganida? (3) 10% qolganida? (4) Nolga aniq yetib bo'lmaydimi?**

[Manba: 2-hafta, amaliy mashg'ulot, topshiriq 13]

Yechim. $(1) E(10)/E_0 = e^{-0.25} \approx 0.7788$: taxminan 77.9%. $(2) e^{-0.025t} = \frac{1}{2} \Rightarrow t = \ln \frac{2}{0}.025 \approx 27.7$ soat.

$(3) e^{-0.025t} = 0.1 \Rightarrow t = \ln \frac{10}{0}.025 \approx 92$ soat.

$$E_0 e^{-0.025t} \approx 92.1 soat. (4) Eksponental arha qanday cheklituchun qat'iy musbat$$

; ular faqat $t \to \infty$ chekligida nolga yaqinlashadi — bu chekli (limit) tushunchasining birinchi tanishuvi.

14. Aholi o'sishi: $P(t) =$ **800e**$^{0.35t}, t$ **soatda. (1) Boshlang'ich aholi soni? (2) 5 soatdan keyin? (3) Ikki barobar bo'lish vaqti? (4) model cheksiz vaqtgacha real bo'lishi mumkinmi?**

[Manba: 2-hafta, amaliy mashg'ulot, topshiriq 14]

Yechim.$_{0ln.35}(1) P(0) = 800$. $(2) P(5) =$ 800e $\approx 800 \cdot 5.755 \approx 4603$. $(3) T = \approx 1.98$ soat. (4) Yo'q: eksponensial o'sish ozuqa moddalari tugashi va joy^2 cheklanishini e'tiborsiz qoldiradi; model haqiqatan faqat madaniyat boshlanishiga yaqin ishlaydi.

3.3 Matematika I: 2-hafta mustaq ish uchun vazifalar — savollar va to'liq yechimlar

1. Har bir munosabat $ynix$ **ning funksiyasi sifatida belgilaydimi:** $a.$ $y =$ **3x − 2;** $b.$ $x^2 + y^2 = 9$**;** $c.$ $y^2 = x + 1$**;** $d.$ $x = y^3 - 2$**.**

[Manba: 2-hafta, ISA, topshiriq 2]

Yechim. $a.$ Funksiya. $b.$ Funksiya emas (aylana: har bir x uchun ikkita y). $c.$ $y = \pm\sqrt{x}+1$: $x > -1$ da ikkita qiymat, funksiya emas. $d.$ $\sqrt[3]{x}+2$ bo'yicha yechamiz; kub ildiz har bir haqiqiy y uchun yagona: $y = x$, shuning uchun bu funksiya (vertikal to'g'ri chiziq testidan o'tadi, gorizontal esa o'tmaydi).

2. $f(x) =$ **2x**$^2-$ **3x** $+1$ **funksiya uchun** $f(0),\ f(-2),\ f(a),\ f(a+h)$**, .** h **ni hisoblang.**$^{f(a+h)-f(a)}$

[Manba: 2-hafta, ISA, topshiriq 3]

Yechim. Amaliy mashg'ulot 2-topshiriq bilan bir xil: 1; 15; 2a$^2-$ 3a $+1$; 2a$^2+$ 4ah $+$ 2h$^2-$ 3a $-$ 3h $+1$; farq nisbati = 4a + 2h − 3.

$$(f(a+h) - f(a))/h = 4a + 2h - 3.$$

3. Tabiiy aniqlanish sohasini toping: $a.$ $f(x) =^{x-1} q$

$$\sqrt{(x+2)/(x-1)}$$

$c.$ $h(x) =_{x-4}$**;** $d.$ $p(x) = \frac{x+2}{x-1}$**.**

[Manba: 2-hafta, ISA, topshiriq 4]

Yechim. $a.$ $(-\infty, \infty)$. $b.$ $(-\infty, \frac{5}{2}].c.x - 1 \geq 0 va x \neq 4$ bo'lishi kerak: $[1, 4) \cup (4, \infty)$. $d.$ $\frac{x+2}{x-1} \geq 0 va x \neq 1$ bo'lishi kerak: kritik nuqtalar $-2 va 1$; kasr manfiy bo'lmagan $(-\infty, -2] \cup (1, \infty)$ oraliqda.

4. $f(x) = x^2 + 2$, $g(x) = -|x-1| + 4$, $h(x) = \sqrt{x} + 3 - 2$ **funksiyalarining qiymatlar to'plamini toping.**

[Manba: 2-hafta, ISA, topshiriq 5]

Yechim. $x^2 \geq 0 \Rightarrow f \geq 2$: qiymatlar to'plami $[2, \infty)$. $|x-1| \geq 0 \Rightarrow -|x-1| \leq 0 \Rightarrow g \leq 4$: $(-\infty, 4]$.$\sqrt{x} + 3 \geq 0 \Rightarrow h \geq -2 : [-2, \infty)$. Har bir javob ichki funksiyaning ekstremal qiymatidan kelib chiqadi.

5. $f(x) = x^2 - 1$, $g(x) = \sqrt{x} + 2$ **bo'lsin.** $f+g, f-g$**, fg,** $f/gvahar$ **bir yangi aniqlanish sohasini aniqlang.**

[Manba: 2-hafta, ISA, topshiriq 6]

Yechim. $(f \pm g)(x) = x^2 - 1 \pm \sqrt{x} + 2$; $(fg)(x) = (x^2-1)\sqrt{x} + 2$; $(f/g)(x) =$

$$(f+g)(x) = x^2 + \sqrt{x} + 1; (f-g)(x) = x^2 - \sqrt{x} - 3;$$
$$(fg)(x) = (x^2-1)\sqrt{x} + 2; (\frac{f}{g})(x) = (x^2-1)/\sqrt{x} + 2.$$

$\sqrt{x+2}$. Sohalar: $f+g, f-g$, fg: $x \geq -2(Rva[-2, \infty)$ ning kesishmasi), ya'ni $[-2, \infty)$. f/g uchun qo'shimcha ravishda $\sqrt{x} + 2 \neq 0$, ya'ni $x \neq -2$: $(-2, \infty)$.

6. $f(x) =$ **2x** -3, $g(x) = x^2 + 1$ **bo'lsin.** $(f \circ g)(x)$, $(g \circ f)(x)$, $(f \circ f)(x)$, $(g \circ g)(x)$ **ni toping.**

[Manba: 2-hafta, ISA, topshiriq 7]

Yechim. $(f \circ g)(x) = 2x^2 - 1$; $(g \circ f)(x) = 4x^2 - 12x + 10$; $(f \circ f)(x) = 2(2x-3) - 3 = 4x - 9$; $(g \circ g)(x) = (x^2+1)^2 + 1 = x^4 + 2x^2 + 2$. Birinchi ikkitasi farqli, shuning uchun kompozitsiya kommutativ emas; oxirgi ikkitasi o'z-o'ziga kompozitsiya to'liq qonuniy ekanligini ko'rsatadi.

7. Har bir funksiyani juft, toq yoki umumiy ko'rinishda ekanligini aniqlang: $x^4 -$ **2x**^{2}**;** $x^3 -$ **4x;** $x^3 + x^2$**;** $|x| + 1$**;** $\frac{x2}{x+1}$**.**

[Manba: 2-hafta, ISA, topshiriq 8]

Yechim. $x^4 - 2x^2$: faqat juft darajalar — juft. $x^3 - 4x$: faqat toq darajalar — toq. $x^3 + x^2$: aralash — umumiy ko'rinishda. $|x| + 1$: juft

$$x^2/(x+1)$$

(amaliy 5-masala). $g(-x) = x^2/(-x+1)$: surat juft, maxraj belgisini o'zgartiradi: $g(-x) \neq \pm g(x)$ — umumiy ko'rinishda.

$\surd$

8. Tasniflang: 3x − 7; $x^4 -$ **2x** $+1$**;** $^{x+1}x^2 + 1$**;** 2^x**;** $\sin x$**.** $_{x2+4}$**;**

$$(x^4 - 2x + 1)/(x+1)$$

[Manba: 2-hafta, ISA, topshiriq 9]

Yechim. 3x − 7: chiziqli (ko'phad). ko'phad. $_{x2+4}$: $\sqrt{x^2} + 1$: algebraik (ildiz). 2^x: eksponensial — ratsional. transsendent. $\sin x$: trigonometrik — transsendent.

9. Amaliy mashq 10 da berilgan chiziqli harorat modeli — barcha besh qismi.

[Manba: 2-hafta, ISA, topshiriq 10]

Yechim. Amaliy mashq 7 dagi model va javoblar bilan bir xil: $T(t) = 20 +$ 4t; burchak koeffitsienti = qizdirish tezligi $4^\circ C/$ min, siljish = boshlang'ich $20^\circ C$; $T(8) = 52^\circ C$; $60^\circ Ct = 10$ min da; soha [0, 15].

10. $p(x) = x^3 -$ **4x va** $r(x) = \frac{x+1}{x-1}$ **uchun aniqlanish sohasini, kesishish nuqtalarini, simmetriyani, chekka xatti-harakatlarini, aniqlanmagan nuqtalarni aniqlang.**

[Manba: 2-hafta, ISA, topshiriq 12]

Yechim. p: amaliy mashq 8 dagi kabi (aniqlanish sohasi R; kesishish nuqtalari -2, 0, 2; toq; chekalarda $\pm\infty$; yo‘q). r: aniqlanish sohasi $x \neq 1$, ya’ni $(-\infty, 1) \cup (1, \infty)$. Kesishish nuqtalari: $r(0) = -1$; surat $x = -1$ da nolga aylanadi. Simmetriya: $r(-x) = \frac{-x+1}{-x-1} = x_{x+1}^{-1} \neq \pm r(x)$ — umumiy ko‘rinishda. Chekka xatti-harakatlar: $r(x) \to 1x \to \pm\infty da$ (gorizontal asimptota $y = 1$); $r(x) \to \pm\infty x \to 1da$ (vertikal asimptota $x = 1$). Faqat $x = 1$ da aniqlanmagan.

11. Mashq 9 amaliyot darsidagi elektr energiyasining qismlarga bo’lingan tariflari — barcha beshta qismi.

[Manba: 2-hafta, ISA, topshiriq 13]

Yechim. Mashq 9 amaliyot darsidagi javoblar bilan bir xil: $C(150) = 15$, $C(200) = 20$, $C(350) = 42.5$; uzluksiz qirrali grafik; qiyaliklar — chekka narxlar; cheklov — 200 kVt·ch chegarasida stimul uzilishi.

12. 30°, 135°, 300° **ni radianga va** $\frac{\pi}{6}$, $\frac{5\pi}{4}$, $\frac{7\pi}{3}$ **ni gradusga aylantiring.**

[Manba: 2-hafta, ISA, topshiriq 14]

Yechim. $\pi/180$ ko’paytiramiz: $30^\circ = \frac{\pi}{6}$; $135^\circ = \frac{3\pi}{4}$; $300^\circ = \frac{5\pi}{3}$. $180/\pi$ ko’paytiramiz: $\frac{\pi}{6} = 30^\circ$; $\frac{5\pi}{4} = 225^\circ$; $\frac{7\pi}{3} = 420^\circ$.

13. O’zaro bir qiymatli bo’ladimi? *a.* $f(x) =$ **3x** $-5, x \in R$**;** *b.* $g(x) = x^2, x \in R$**;** *c.* $h(x) = x^2, x \geq 0$**;** *d.* $p(x) = \frac{1}{x}$, $x \neq 0$**.**

[Manba: 2-hafta, ISA, topshiriq 15]

Yechim. *a.* Ha: qiyalik chiziq gorizontal chiziq testidan o’tadi. *b.* Yo’q: $g(-2) = g(2) = 4$. *c.* Ha: $[0, \infty)$ da kvadratga olish injektivdir (cheklov gorizontal chiziq testini tiklaydi). *d.* Ha: $\frac{1}{x1} = \frac{1}{x2} \Rightarrow x_1 = x_2$.

14. Ters funksiyalari va ularning aniqlanish sohalarini toping: *a.* $f(x) =$ **4x − 7;** *b.* $g(x) = \frac{x+3}{2}$**;** *c.* $h(x) = x^2 + 1$, $x \geq 0$**;** *d.* $p(x) = \frac{2x+3}{x-1}$, $x \neq 1$**; kompozitsiya orqali tekshiring.**

[Manba: 2-hafta, ISA, topshiriq 16]

Yechim. *a.* $f^{-1}(x) = (x + 7)/4$, aniqlanish sohasi R (amaliyot 10). *b.* $y = (x + 3)/2 \Rightarrow$

$$y = (x + 1)^2 \Rightarrow x = \sqrt{y - 1}$$

$x =$ 2y − 3, demak $g^{-1}(x) =$ 2x − 3, aniqlanish sohasi R. *c.* (musbat ildiz, chunki $x \geq 0$): $h^{-1}(x) = \sqrt{x - 1}$, aniqlanish sohasi $[1, \infty)$. *d.* $y = (2x + 3)/(x - 1) \Rightarrow y(x - 1) =$ 2x +3 $\Rightarrow x(y - 2) = y + 3 \Rightarrow x = (y + 3)/(y - 2)$: $p^{-1}(x) = (x + 3)/(x - 2)$, aniqlanish sohasi $x \neq 2$. Har biri qurilish bo’yicha $f(f^{-1}(x)) = x$ ni qanoatlantiradi.

15. A, B, C, D **uchun o’sish/yemirilish klassifikatsiyasi (amaliyot 11).**

[Manba: 2-hafta, ISA, topshiriq 17]

Yechim. Xuddi shu: A o’sish $(a = 1.08)$, B yemirilish $(a = 0.85)$, C o’sish $(k = 0.03)$, D yemirilish $(k = -0.4)$.

16. Yeching, avval aniq javoblar beramiz: *a.* $2^{x+1} = 16$**;** *b.* $5^2x = 125$**;** *c.* $e^3x = 7$**;** *d.* $\mathbf{100e}^{-0.2t} = 20$**.**

[Manba: 2-hafta, ISA, topshiriq 18]

Yechim. $a.\ x+1=4 \Rightarrow x=3.\ b.\ 2\mathrm{x}=3 \Rightarrow x=\frac{3}{2}.\ c.\ 3\mathrm{x}=\ln 7 \Rightarrow x=\ln^{7} \approx 0.6486.\ 3d.$ $e^{-0.2t}=0.2 \Rightarrow t=-^{ln0.2}=5\ln 5 \approx 8.047.\ 0.2$

17. Batareyaning zaryadi (amaliy mashq 13) — barcha to'rt qismi.

[Manba: 2-hafta, ISA, topshiriq 19]

Yechim. Xuddi shu: $e^{-0.25} \approx 77.9\%$ 10 soatdan keyin; 50% $\frac{ln2}{0.025} \approx 27.7$ soatda; 10% $\frac{ln10}{0.025} \approx 92.1$ soatda; eksponensiya chekli vaqtda nolga yetmaydi.

18. Bakteriya populyatsiyasi (amaliy mashq 14) — barcha to'rt qismi.

[Manba: 2-hafta, ISA, topshiriq 20]

Yechim. Xuddi shu: $P_0 = 800$; $P(5) \approx 4603$; ikkilanish vaqti $\frac{ln2}{0.35} \approx 1.98$ soat; sof eksponensial modellar uzoq muddatli istiqbolda (resurslar, joy, chiqindilar) muvaffaqiyatsizlikka uchraydi.

4 3-hafta — Chegaralar (Math I)

4.1 Math I: 3-hafta formulalar katalogi

4.1.1 O'zgarish tezliklari va ikki asoschi muammo

Chegaralar kursga ikki savol orqali kirib keladi: kattalik qanchalik tez o'zgarayotgani (urinma/tezlik) va funksiya qiymatlari nuqta yaqinida qanday xulq atayotgani. Qisqarayotgan oraliqdagi o'rtacha tezlik oniy tezlikni aniqlaydi.

$$m_{\text{sec}} = \frac{f(x)-f(a)}{x-a} \ (\text{(}a, f(a)\text{)} \text{ dan o'tuvchi sekantning qiyaligi}) \tag{51}$$

$$v_{\text{ave}} = \frac{s(t)-s(a)}{t-a} \ (\text{[}a,t\text{]} \text{ oraliqdagi o}\prime\text{rtacha tezlik}) \tag{52}$$

$$\lim_{h\to 0}\frac{f(a+h)-f(a)}{h} = f'(a) \text{ (oniy tezlik; limit sifatida aniqlanadi)} \tag{53}$$

4.1.2 Limit tushunchasi, bir tomonlama va cheksiz chegaralar

Intuitiv ta'rif (OpenStax §2.2) va uning bir tomonlama variantlari haftaning amaliy yuragi; ikki tomonlama limit aynan ikkala bir tomonlama limit mos kelganda mavjud.

$$\lim_{x\to a} f(x) = L \Leftrightarrow \text{barcha } f(x)\ (x\neq a)\ x\to a \text{ da } L \text{ ga intiladi} \tag{54}$$

$$\lim_{x\to a^-} f(x) = L, \quad \lim_{x\to a^+} f(x) = L \text{ (chap/o'ng chegaralar)} \tag{55}$$

$$\lim_{x\to a} f(x) = L \Leftrightarrow \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L \text{ (mavjudlik mezoni)} \tag{56}$$

$$\lim_{x\to a^-} f(x) = \pm\infty, \quad \lim_{x\to a^+} f(x) = \pm\infty, \quad \lim_{x\to a} f(x) = \pm\infty \text{ (cheksiz chegaralar)} \tag{57}$$

$$\lim_{x\to a^-} \frac{1}{(x-a)^n} = +\infty, \quad \lim_{x\to a^+} \frac{1}{(x-a)^n} = +\infty \quad (n \text{ juft}) \tag{58}$$

$$\lim_{x\to a^-}\frac{1}{(x-a)^n}=-\infty,\quad \lim_{x\to a^+}\frac{1}{(x-a)^n}=+\infty\quad (n \text{ toq}):\ x=a \text{ vertikal asimptota} \tag{59}$$

$x=a$ to'g'ri chizig'i f ning vertikal asimptotasi, agar (57) dagi bir tomonlama cheksiz chegaralarning kamida bittasi o'sha yerda bajarilsa.

4.1.3 Aniq ta'rif (§2.5, ma'ruzada ta'kidlangan)

$\varepsilon-\delta$ ta'rifi «ixtiyoriy darajada yaqin» ni miqdorlashtirilgan masofalarga aylantiradi; ekvivalent modul tengsizliklar — ish vositasi.

$$\lim_{x\to a}f(x)=L \Leftrightarrow \forall\varepsilon>0\ \exists\delta>0:\ 0<|x-a|<\delta \Rightarrow |f(x)-L|<\varepsilon \tag{60}$$

$$|f(x)-L|<\varepsilon \Leftrightarrow L-\varepsilon<f(x)<L+\varepsilon \tag{61}$$

$$0<|x-a|<\delta \Leftrightarrow a-\delta<x<a+\delta \text{ va } x\neq a \tag{62}$$

Bu **haftada ko'rilgan ikki standart chegara**

$$\lim_{x\to 0}\frac{\sin x}{x}=1 \text{ (ma/ruza 2.2-jadvali: qiymatlar } 0.9983,\ 0.99998,\ 0.9999998\ \ldots) \tag{63}$$

$$\lim_{x\to 0}\sin\frac{1}{x} \text{ mavjud emas (tebranish)},\quad \lim_{x\to 0^-}\frac{1}{x}=-\infty,\quad \lim_{x\to 0^+}\frac{1}{x}=+\infty \tag{64}$$

4.2 Math I: 3-hafta ISA — savollar va to'liq yechimlar

1. Dars va amaliy mashqda taqdim etilgan barcha masalalarni, mashqlarni va misollarni o'rganing; tez va xato qilmasdan javob bera olishingiz kerak.
[Manba: 3-hafta, ISA, topshiriq 2]

Yechim. Bu o'rganish ko'rsatmasi, hisoblash emas; tegishli material to'liq dars misollarida, pastdagi amaliy mashq blokida (mashqlar 2–8) va tayinlangan kitob masalalarida (keyingi bandlarida) yechilgan.

2. $f(x)=x^3$ **uchun** $x=1$ **da tegishish burchak koeffitsientini** $m_{sec}=f(1+h)-f(1)$

$$m_{sec}=3+3h+h^2$$

yordamida baholang, chap va o'ngdan bir xil songa intiladimi?
[Manba: 3-hafta, ISA, topshiriq 3]

Yechim. Bu yerda $f(1)=1 va f(1+h)=(1+h)^3=1+$ 3h + $3h^2+h^3$, shuning uchun $m_{sec}=3+$ 3h $+h^2$. Qiymatlar: $h=-0.1\Rightarrow 2.71$; $-0.01\Rightarrow 2.9701$; $-0.001\Rightarrow 2.997$; $0.001\Rightarrow 3.003$; $0.01\Rightarrow 3.0301$; $0.1\Rightarrow 3.31$. Ikkala tomon ham 3 ga yaqinlashadi; aslida $\lim_{h\to 0}(3+3h+h^2)=3$ — bu egri chiziqqa urinma aniq burchak koeffitsiyenti (hosila $3x^2 dax=1da$).

3. Open-Stax Calculus 1 dagi "1-bob" (mashqlar ro'yxati §2.2) dagi 46-$49 va 60$**-64-masalalarni yeching.**
[Manba: 3-hafta, ISA, topshiriq 4]

Yechim. Bu grafiklarni o'qish bo'yicha mashqlar §4.4 da yozib qo'yilgan va to'liq yechilgan; kurs varag'idagi raqamlash aynan shu mashqlar ro'yxatiga tegishli.

4. Temperatura regulyatori $T < 50 da C(T) = 0 va T \geq 50 da C(T) = 10$ **(volt) beradi.** $\lim_{T} \to 50 - C(T)$, $\lim_{T} \to 50 + C(T)$, $\lim_{T} \to 50 C(T)$ **ni aniqlang; operatsiyani o'tkazish sifatida talqin qiling.**

[Manba: 3-hafta, ISA, topshiriq 5]

Yechim. Chapda regulyator "o'chirilgan" shoxni ko'rsatadi: $\lim_{T} \to 50 - C(T) = 0$. O'ngda "yoqilgan": $\lim_{T} \to 50 + C(T) = 10$. Bir tomonlama limitlar har xil bo'lgani uchun, ikki tomonlama limit mavjud emas (56). Talqin: $T = 50$ da keskin o'tkazish — chiqish $0V dan 10V$ ga sakrab o'tadi, aynan shunday termostat isitgichni boshqaradi.

$$s(t) = 2t^2 + 3t (m, s). (1)$$

5. Robot-aravachaning holati: (1) O'rtacha tezlik $s(2 + h) - s(2)$

$$s(2 + h) - s(2) = (2(2 + h)^2 + 3(2 + h)) - 14$$

$(3) \lim_{h \to 0}$ **ni hisoblang; (4) talqin qiling.** h

$$s(3) - 14 = (27 - 14)/(14.72 - 14)$$

[Manba: 3-hafta, ISA, topshiriq 6]

Yechim. $(1) s(2) = 14$. Qiymatlar: $h = 1 : = 13$; $h = 0.1 : =$

$$s(2 + h) - s(2)/h = (14.07(2 + h)^2 - 14)/(h) = (14.07(4 + 4h + h^2) - 14)/(h) =$$

$$(56.28 + 56.28h + 14.07h^2 - 14)/(h) = (42.28 + 56.28h + 14.07h^2)/(h)$$
$$42.28/h + 56.28 + 14.07h (9)$$

7.2; $h = 0.01 : \ ^{14.0702-14} = 7.02$; $h = -0.01 : \ ^{13.9302-14} = 6.98$. $(2) s(2 + h) = 2(2 + h)^2 + 3(2 +$

$$s(2 + h) - s(2) = 2(2 + h)^2 + 3(2 + h) - (2 \cdot 2^2 + 3 \cdot 2) = 8h + 2h^2, (2)$$

$h) = 8 + \ 8\text{h} + 2\text{h}^2 + 6 + \ 3\text{h} = 14 + \ 11\text{h} + 2\text{h}^2$, shuning uchun $^{s(2+h)-s(2)} == 11 + \ 2\text{h}$. $(3) hh \lim \frac{11h+2h2}{h \to 0} (11 + 2h) = 11m/s$. (4) Aravachaning $t = 2s$ dagi lazer tezligi $11m/s$; o'rtacha qiymatlar ikkala tomondan ham unga yaqinlashadi.

6. Ikkala sistema chiqishi $A(t) = (1.08)^t va B(t) = e^{0.03t}$**.** $(1) \lim_{t \to \infty} A(t) va \lim_{t \to \infty} B(t)$ **ni hisoblang;** $(2) A(t) nie^{kt}$ **ko'rinishida yozing; (3) qaysi tezroq o'sadi?** $(4) \lim_{t \to \infty} A(t)$**}** $\frac{B}{(t)}$ **} ni hisoblang; (5) talqin qiling.**

[Manba: 3-hafta, ISA, topshiriq 7]

Yechim. (1) Ikkala asos ham 1 dan katta, shuning uchun ikkala limit ham $+\infty$. $(2) A(t) = e^{tln 1.08}$, bunda $\ln 1.08 \approx 0.0770$, demak $A(t) = e^{0.0770t}$. $(3) A$ tezroq o'sadi, chunki uning tezlik doimiysi $0.0770 > 0.03$. $(4) \frac{B(t)}{A(t)} = e^{(0.03-0.0770)t} = e^{-0.047t} \to 0t \to \infty$ da. (5) Ikkalasi ham cheksizlikka intilsa ham, ularning nisbati nolga intiladi: BA ga nisbatan ahamiyatsiz bo'lib qoladi — 4-hafta "kichik o" taqqoslashining boshlanishi.

4.3 Matematika I: 3-hafta amaliy mashg'uloti (muhandislik limitlari) — savollar va to'liq yechimlar

Bu mashg'ulot uchun kurs talablari: matematik jarayonni ko'rsatish; birliklarni kiritish; muhandislik atamalarida talqin qilish; limit mavjud bo'lmaganda ko'rsatish.

1. Darslikning §2.2 mashqlar ro'yxatidagi 50-54-masalalarni yeching.
[Manba: 3-hafta, amaliy mashg'ulot, mashq 1]

Yechim. Yozib qo'yilgan va to'liq yechilgan $§4.4da(50-54-$bandlar).

2. Bir tomonlama limitlar: $f(x) =$ **2x**$+1x < 2da,\ f(x) = 7 - xx \geq 2$ **da.** $\lim_{x\to 2} -f$, $\lim_{x\to 2} +f,\ \lim_{x\to 2} f,\ f(2)$ **ni toping; ikki tomonlama limit mavjudligini tushuntiring.**
[Manba: 3-hafta, amaliy mashg'ulot, mashq 2]

Yechim. Chap shox: $\lim_{x\to 2} -(2x+1) = 5$. O'ng shox: $\lim_{x\to 2} +(7-x) = 5$. Ular teng, demak (56) bo'yicha $\lim_{x\to 2} f(x) = 5$. E'tibor bering, $f(2) = 7-2 = 5$ ham — funksiya qiymati limitga teng, shuning uchun $x = 2$ da grafikda uzilish yo'q.

$$V(T) = (2500/T)MB, T \geq 50$$

3. Kalibrovka yaqinidagi o'lchovsizlik: $V(T) =$ **mV,** $T \neq 50T - 50(T^{\circ}Cda)$**. (1) Nima uchun** $V(50)$ **aniqlanmagan?** $(2)\lim_T \to 50V(T)$ **ni baholang; (3) limit qanday yo'qolgan kalibrovka qiymatini almashtira oladi?**
[Manba: 3-hafta, amaliy mashg'ulot, 3-mashq]

Yechim. (1) Formula $T-50$ ga bo'ladi, bu $T = 50da0$ ga teng; sensor modelining shu nuqtada qiymati yo'q. (2) Kvadratlar ayirmasini ajratamiz: $T^2 - 2500 = (T-50)(T+50)$, demak $T \neq 50da, V(T) = T + 50$, va $\lim_T \to 50V(T) = 100$ mV. (3) limit sensor $50^{\circ}C$ da ko'rsatadigan narsani tasvirlaydi; kalibrovka jadvalining bu nuqtada 100 mV qiymati bilan to'ldirish to'g'ri, chunki chiqish qiymati ikki tomondan ham bu qiymatga yaqinlashadi — bu "olib tashlanadigan" xulq-atvor, emaski sakchiq.

4. O'zgaruvchi tizim: $V(t) =$ **2t** $+1,\ t < 4vaV(t) = 12 - t,\ t \geq 4(V)$**. Bir tomonlama limitlarni, ikki tomonlama limitni va** $V(4)$ **ni aniqlang;** $t = 4$ **da kuchlanishda keskin o'zgarish bormi?**
[Manba: 3-hafta, amaliy mashg'ulot, 4-mashq]

Yechim. Chapdan: $\lim_{t\to 4} -(2t+1) = 9V$. O'ngdan: $\lim_{t\to 4} +(12-t) = 8V$. $V(4) = 12-4 = 8V$. Bir tomonlama limitlar har xil, demak $\lim_{t\to 4} V(t)$ mavjud emas: $t = 4$ da quvvat $9Vdan8V$ ga keskin pasayadi. Muhandislik ma'nosi: $1V$ qadam boshqariladigan qurilmada o'tish jarayonlarini keltirib chiqarishi mumkin; qadam zararli bo'lganda, kirish filtrlari yoki yumshoq ishga tushirish mantig'i qo'llaniladi.

5. Kuchlanish-deformatsiya: $\sigma(\varepsilon) = 200000\varepsilon,\ \varepsilon < 0.002va\sigma(\varepsilon) = 400 + 10000(\varepsilon - 0.002),\ \varepsilon \geq 0.002(\sigma MPada;$ **Gyuk sohasi, keyi mustahkamlanish).** $\varepsilon = 0.002da$ **bir tomonlama limitlarni toping; ikki tomonlama limit mavjudmi? Model sakchiq bashorat qiladimi? O'tish yaqinidagi grafikni chizing.**
[Manba: 3-hafta, amaliy mashg'ulot, 5-mashq]

Yechim. Elastik tomon: $\lim_{\varepsilon\to 0.002} -200000\varepsilon = 200000(0.002) = 400$ MPa.

$$\lim_{\varepsilon\to 0.002} -\sigma(\varepsilon) = 200000(0.002) = 400MPa;$$
$$\lim_{\varepsilon\to 0.002} +\sigma(\varepsilon) = 400 + 10000(0.002 - 0.002) =$$

$$400MPa; \lim_{\varepsilon \to 0.002} \sigma(\varepsilon) = 400MPa(mavjud).$$

Plastik tomon: $400 + 10000(\varepsilon - 0.002) = 400$ MPa. Ikkalasi ham 400 MPa ga teng, demak ikki tomonlama limit mavjud va 400 MPa ga teng; sakchiq yo‘q — bo‘lakli model oqim nuqtasida uzluksiz, faqat qiyalik o‘zgaradi (200 000 dan 10 000 MPa gacha birlik deformatsiyasiga). Grafik — (0.002, 400) nuqtada uchrashadigan ikki to‘g‘ri chiziq bo‘lagi, aniq burilish bilan: elastik yuklanish, keyin ancha tekisroq plastik shox.

6. RC**-zanjirining uzoq muddatli javobi:** $V(t) = V_s(1 - e^{-t/(RC)})$, $V_s = 12V$, $R = 10000\Omega$, $C = 100$ **µ**F**.** $(1)\lim_{t\to 0+} V(t)$ **ni hisoblang;** $(2)\lim_{t\to\infty} V(t)$**; (3) ikkalasini ham fizik talqin qiling.**

[Manba: 3-hafta, amaliy mashg‘ulot, 6-mashq]

Yechim. $(1)t = 0^+$ da eksponensial $e^0 = 1$ ga teng, demak $V(0^+) = 12(1-1) = 0$. $(2)t \to \infty da, e^{-t/(RC)} \to 0, V$ — boshlang‘ichda zaryadsiz kondensator. Demak $V \to V_s = 12V$. (3) Kondensator $0Vdan12V$ quvvatlanish kuchlanishiga zaryadlanadi, unga asimptotik yaqinlashadi; vaqt doimiysi $\tau = RC = 10^4 \cdot 10^{-4} = 1s$ tezlikni belgilaydi (taxminan $1sda63\%$ zaryad, $5\tau da99\%$). ”$V \to V_s t \to \infty$ da” limit tili — bu aynan asimptotik zaryadlanishni ifodalash usuli.

7. Doimiy tezlik: $v(t) = v_T(1 - e^{-kt})$, $v_T > 0$, $k > 0$**.** $(1)v(0)$**?** $(2)\lim_{t\to\infty} v(t)$**? (3) Nima uchun bu cheksiz tezlanishdan realroq model?**

[Manba: 3-hafta, amaliy mashg‘ulot, 7-mashq]

Yechim. $(1)v(0) = v_T(1 - 1) = 0$: jism tinch holatdan boshlanadi. $(2)e^{-kt} \to 0$, demak $v \to v_T$: tezlik doimiy holatga yaqinlashadi. (3) Havo qarshiligi tezlik bilan oshadi, tortishish kuchiga teng bo‘lguncha; shundan keyin umumiy kuch nolga teng va tezlik oshaverishi mumkin emas — tushuvchi jism haqiqatan ham chekli limitga yaqinlashadi, cheksiz tezlanmaydi.

8. Ma’lumot uzatish tezligi: $R(n) =^{100} n$ **Mbit**$/s$**.** $(1)n + 20R(10)$, $R(100)$, $R(1000)$ **ni hisoblang;** $(2)\lim_{n\to\infty} R(n)$**;** $(3)n$ **ni oshirish nima uchun oxir-oqibat kamroq yaxshilanishlarni beradi? (4) maksimal nazariy tezlik?**

[Manba: 3-hafta, amaliy mashg’ulot, mashq 8]

Yechim. $(1)R(10) \approx 33.3$, $R(100) \approx 83.3$, $R(1000) \approx 98.0$ Mbit$/s$. (2) ni bo’lganda: $R(n) = \frac{100}{1+20/n} \to 100$ Mbit$/s$. (3) Har bir qo’shimcha resurs birligi tobora katta asosga bo’linadi, shuning uchun chekli foyda kamayadi — kamayadigan samaradorlik. (4) Gorizontal asimptota 100 Mbit$/s$ — modelning limiti: unga yaqinlashadi, lekin undan oshmaydi.

4.4 Matematika I: darslik tayinlangan mashqlari (OpenStax §2.2, bandlar 46–54 va 60–64)

Bu grafiklarni o’qish mashqlari to’plam bilan birga chop etilgan to’rtta rasmni o’z ichiga oladi; kerakli qiymatlar, auditoriyada bo’lgani kabi, grafiklardan olinadi. Avval shartlar, keyin yechimlar.

1. Birinchi grafik $y = f(x)$ **uchun yeching:** $(46)\lim_{x\to 10} -f(x) = 0$**;** $(47)\lim_{x\to -2} +f(x) = 3$**;** $(48)\lim_{x\to -8} f(x) = f(-8)$**;** $(49)\lim_{x\to 6} f(x) = 5$ **— to’g’ri yoki noto’g’ri?**

[Manba: OpenStax §2.2, mashq 46–49]

Yechim. (46) To'g'ri: $x \to 10$ chap tomondan egri chiziq bo'yalgan nuqtaga (10, 0) tushadi, ya'ni chap limit 0. (47) To'g'ri: -2 biroz o'ng tomonda grafik 3 balandlikda joylashgan (bo'yalgan nuqta (-2, 3)). (48) Noto'g'ri: $x = -8$ yaqinida egri chiziq -6 balandlikdagi ochiq doira yaqiniga yaqinlashadi, ya'ni $\lim_{x\to -8} f(x) = -6, f(-8) = -3$ (bo'yalgan nuqta) esa; limit va qiymat bir xil bo'lmasligi mumkin. (49) Noto'g'ri: chap tomondan egri chiziq ochiq doiraga (6, 2) tushadi, ya'ni $\lim_{x\to 6} - = 2$, o'ng shakchi esa bo'yalgan nuqtadan (6, 5) chiqadi, ya'ni $\lim_{x\to 6} + = 5$; bir tomonlama limitlar har xil $\Rightarrow$ ikki tomonlama limit mavjud emas, shuning uchun $u5$ ga teng bo'lishi mumkin emas.

2. Qismchali chiziqli grafik uchun $(y = x + 1$ to'g'ri chizig'i ochiq nuqtagacha (1, 2); bo'yalgan nuqta (1, 1); keyin $y = 3 - x$ to'g'ri chizig'i), toping $(50)\lim_{x\to 1} -f$; $(51)\lim_{x\to 1} +f$; $(52)\lim_{x\to 1} f$; $(53)\lim_{x\to 2} f$; $(54) f(1)$.

[Manba: OpenStax §2.2, mashq 50–54]

Yechim. (50) Chap shoxi $y = x + 1 \to 2$. (51) O'ng shoxi $y = 3 - x \to 2$. (52) Ikkala tomon ham 2 ga teng, ya'ni $\lim_{x\to 1} f(x) = 2$ — hatto $f(1) = 1$ bo'lsa ham; limit o'z nuqtani e'tiborsiz qoldiradi. $(53) x = 2$ da faqat o'ng shoxi yashaydi: $3 - 2 = 1$. $(54) f(1) = 1$ (bo'yalgan nuqta).

3. Uchinchi grafik uchun (chap tomonda: ochiq nuqtagacha (0, 1) tushadigan to'g'ri chiziq; o'ng tomonda: bo'yalgan nuqtadan (0, -4) chiqib, $x = 2$ yaqinida nolni kesib o'tuvchi egri chiziq), toping $(55)\lim_{x\to 0} -f$; $(56)\lim_{x\to 0} +f$; $(57)\lim_{x\to 0} f$; $(58)\lim_{x\to 2} f$.

[Manba: OpenStax §2.2, mashq 55–58]

Yechim. (55) 1. (56) -4. $(57) 1 \neq -4$, shuning uchun ikki tomonlama limit mavjud emas (56). (58) O'ng shoxi $x = 2$ da o'qni kesib o'tadi, ya'ni limit 0.

4. To'rtinchi grafik uchun (uch qism: ochiq nuqtagacha (-2, 0) ko'tariladigan to'g'ri chiziq; bo'yalgan nuqtadan (-2, 2) boshlab koordinata boshidan (2, -2) gacha pasayadigan kesma; ochiq nuqtadan (2, 2) boshlab ko'tariladigan to'g'ri chiziq), toping $(59)\lim_{x\to -2} -f$; $(60)\lim_{x\to -2} +f$; $(61)\lim_{x\to -2} f$; $(62)\lim_{x\to 2} -f$; $(63)\lim_{x\to 2} +f$; $(64)\lim_{x\to 2} f$.

[Manba: OpenStax §2.2, mashq 59–64]

Yechim. (59) 0. (60) 2. $(61) 0 \neq 2$: mavjud emas. (62) -2. (63) 2. $(64) -2 \neq 2$: mavjud emas. Ikkala birikish haqiqiy sakrashlar: hech qanday o'rtacha qiymat teng bo'lmagan bir tomonlama limitlarni tuzatmaydi.

5 4-hafta — Chegaralar II: chegaralar qonunlari, standart chegaralar, noaniqliklar, asimptotika, Landau belgilari (Matematika I)

4-hafta — Test 1 uchun takrorlashdan oldin asosiy hafta: $u3$-haftadagi chegaralar apparatini ishlovchi asbobga aylantiradi. Majburiy o'qish — OpenStax §2.3 (Chegaralar qonunlari), prof. Neofotistosning 4-hafta konspekti chegaralarning umumiy xususiyatlarini, standart chegaralarni, noaniqliklarni, asimptotik ekvivalentlikni va Landau $Ovao$ belgilarini umumlashtiradi. Bu qism ikkala manbadagi har bir tenglamani sanab o'tadi, keyin §2.3 to'plamidagi har bir mashqni (bandlar 83–130) va konspektning tahlil qilingan har bir savolini yechadi.

5.1 4-hafta formulalari katalogi

5.1.1 Chegaralarning umumiy xossalari (4-hafta konspekti)

$$\lim_{x\to a} f(x) = L: \ f(x) \text{ ni } L \text{ ga ixtiyoriy darajada yaqinlashtirish mumkin } (x \to a,\ x \neq a) \tag{78}$$

$$\lim_{x\to a} f(x) = L \Leftrightarrow \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L \text{ (ikki tomonlama mezon)} \tag{79}$$

limit, mavjud bo'lganda, yagona.

5.1.2 Chegara qonunlari (OpenStax Teorema 2.5)

Agar $\lim x \to af(x) = Lva \lim x \to ag(x) = M$ (ikkalasi ham mavjud)vac konstanta bo'lsa:

$$\text{Yig/indi: } \lim_{x\to a}[f(x)+g(x)] = L+M; \quad \text{Ayirma: } \lim_{x\to a}[f(x)-g(x)] = L-M \tag{80}$$

$$\text{Konstanta ko/paytma: } \lim_{x\to a} c\,f(x) = cL; \quad \text{Ko/paytma: } \lim_{x\to a}[f(x)g(x)] = LM \tag{81}$$

$$\text{Bo/linma: } \lim_{x\to a}\frac{f(x)}{g(x)} = \frac{L}{M}\ (M\neq 0); \quad \text{Daraja: } \lim_{x\to a}[f(x)]^n = L^n\ (n\in\mathbb{N}) \tag{82}$$

$$\text{Ildiz: } \lim_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{L}\ (n \text{ toq: istalgan } L;\ n \text{ juft: } L\geq 0) \tag{83}$$

$$\text{Asosiy: } \lim_{x\to a} x = a, \quad \lim_{x\to a} c = c \tag{84}$$

$$\text{Ko/phadlar/ratsional funksiyalar: } \lim_{x\to a} p(x) = p(a), \quad \lim_{x\to a}\frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}\ (q(a)\neq 0) \tag{85}$$

Agar a yaqinida $x \neq a$ bo'lgan barcha joyda $f(x) = g(x)$ bo'lsa, u holda $\lim x \to af(x) = \lim x \to ag(x)$ — qo'yishdan oldin ko'paytuvchilarni qisqartirish litsenziyasi.

5.1.3 Siqish (Pinchlovchi) teoremasi (OpenStax Teorema 2.7)

$$f(x)\leq g(x)\leq h(x)\ (a \text{ yaqinida},\ x\neq a), \quad \lim_{x\to a} f(x) = L = \lim_{x\to a} h(x) \Rightarrow \lim_{x\to a} g(x) = L \tag{86}$$

5.1.4 Standart chegaralar va ulardan olinadigan trigonometrik chegaralar (4-hafta konspekti)

$$\lim_{x\to 0}\frac{\sin x}{x} = 1, \quad \lim_{x\to 0}\frac{1-\cos x}{x^2} = \frac{1}{2}, \quad \lim_{x\to 0}\frac{e^x-1}{x} = 1 \tag{87}$$

$$\lim_{x\to 0}\frac{\ln(1+x)}{x} = 1, \quad \lim_{x\to 0}(1+x)^{1/x} = e, \quad \lim_{x\to\infty}\frac{x^n}{e^x} = 0\ (n\in\mathbb{N}) \tag{88}$$

Trigonometrik chegaralarda burchaklar radian bo'lishi shart. (87) ni almashtirish orqali moslashtiramiz, masalan:

$$\lim_{x\to 0}\frac{\sin 5x}{x}=5\lim_{x\to 0}\frac{\sin 5x}{5x}=5,\quad \lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1\Rightarrow\lim_{\theta\to 0}\frac{1-\cos\theta}{\theta}=0 \text{ (qo⁄shma son)} \tag{89}$$

5.1.5 Aniqlanmagan holdalar (4-hafta konspekti)

$$\frac{0}{0},\ \frac{\infty}{\infty},\ 0\cdot\infty,\ \infty-\infty,\ 0^0,\ 1^\infty,\ \infty^0 \tag{90}$$

«Aniqlanmagan» so'zi limit mavjud emas degani emas; bu qo'shimcha tahlil kerak degani. Bir xil boshlang'ich hold har xil natijaga ega bo'lishi mumkin:

$$\lim_{x\to 0}\frac{x}{x}=1,\quad \lim_{x\to 0}\frac{x}{x^2}=0,\quad \lim_{x\to 0}\frac{x^2}{x}\text{ mavjud emas (chekli)} \tag{91}$$

holbuki ko'paytuvchilarga ajratgandan keyin $\lim x\to 1(x^2-1)/(x-1)=2$. Hal qilish vositalari: ko'paytuvchilarga ajratish, ratsionallashuv, standart chegaralar, asimptotik ekvivalentlik, o'zgaruvchini almashtirish, algebraik qayta tuzish, Siqish teoremasi, Taylor yoyilmalari va (keyinroq) L'Hôpital qoidasi.

5.1.6 Asimptotik ekvivalentlik (4-hafta konspekti)

$$f(x)\sim g(x)\ (x\to a)\Leftrightarrow\lim_{x\to a}\frac{f(x)}{g(x)}=1 \tag{92}$$

$$f(x)\sim g(x)\Leftrightarrow f(x)=g(x)\big(1+\varepsilon(x)\big),\ \varepsilon(x)\to 0 \tag{93}$$

$$f\sim g\Leftrightarrow f=g+o(g) \tag{94}$$

$$x\to 0\text{ da: }\sin x\sim x,\ \tan x\sim x,\ 1-\cos x\sim\frac{x^2}{2},\ e^x-1\sim x \tag{95}$$

$$\ln(1+x)\sim x,\quad (1+x)^\alpha-1\sim\alpha x\ (\alpha\text{ konstanta}) \tag{96}$$

Raqamli jihatdan $\sin(0.01)/0.01\approx 0.999983$ «yaqin, lekin teng emas»ni illyustratsiya qiladi. Cheksizlikda eng yuqori darajali had hukmron:

$$3x^2+5x-1\sim 3x^2\ (x\to\infty),\quad \frac{2x^3+x}{5x^2-1}\sim\frac{2x^3}{5x^2}=\frac{2}{5}x\ (x\to\infty) \tag{97}$$

Ekvivalentliklar ko'paytirilishi va bo'linishi mumkin, umuman qo'shilmasligi mumkin.

5.1.7 Landau belgilari: katta-O va kichik-o(4−hafta konspekti)

$$f(x)=O\big(g(x)\big)\ (x\to a)\Leftrightarrow\frac{|f(x)|}{|g(x)|}\ a\text{ yaqinida chegaralangan qoladi} \tag{98}$$

$$f(x)=o\big(g(x)\big)\ (x\to a)\Leftrightarrow\lim_{x\to a}\frac{f(x)}{g(x)}=0 \tag{99}$$

$$f = o(g) \Rightarrow f = O(g); \quad f = O(g) \not\Rightarrow f = o(g) \tag{100}$$

$$x^2 = o(x),\ x^3 = o(x^2)\ (x \to 0) \tag{101}$$

$$x = o(x^2),\ x^{10} = o(e^x)\ (x \to \infty) \tag{102}$$

$$3x = O(x) \text{ lekin } 3x \neq o(x)\ (x \to 0) \text{ chunki } \frac{3x}{x} = 3 \tag{103}$$

$$\sin x = x + o(x)\ (x \to 0); \quad e^x = 1 + x + o(x); \quad e^x = 1 + x + \frac{x^2}{2} + o(x^2) \tag{104}$$

$$2x^2 + 3x + 1 = 2x^2 + O(x)\ (x \to \infty) \tag{105}$$

So'zlarda: katta-O — f ning kattalik tartibi g nikiga ko'pi bilan teng; kichik-o — fg bilan solishtirganda ahamiyatsiz bo'lib qoladi. Ehtiyot bo'ling: $f = O(g)$ oddiy ma'noda tenglama emas — $O(g)$ butun funksiyalar sinfini bildiradi.

5.2 OpenStax §2.3, nazorat masalalari 2.11–2.20 — savollar va to'liq yechimlar

1. Chekimalar qonunlaridan foydalanib, $\lim_{x\to -3}(2x-1)\sqrt{x}+4$ **ni hisoblang, har bir bosqichda qonunni ko'rsating.**
[Manba: OpenStax §2.3, nazorat 2.11]

Yechim. Ikkala ko'paytuvchining cheklari mavjud, shuning uchun ko'paytma qonunini qo'llaymiz: $\lim(2x-1) = 2(-3)-1 = -7$ (yig'indi, ayirma, asosiy chekimalar); $\lim \sqrt{x}+4 = \sqrt{-3}+4 = 1$ (ildiz qonuni). Ko'paytma: $-7 \cdot 1 = -7$.

2. $\lim_{x\to -2}(3x^3 - 2x + 7)$ **ni hisoblang.**
[Manba: Nazorat 2.12]

Yechim. To'g'ri almashtirish asosli (85): $3(-8) - 2(-2) + 7 = -24 + 4 + 7 = -13$.

3. $\lim_{x\to -3}(x^2 + 4x + 3)/(x^2 - 9)$ **ni hisoblang.**
[Manba: Nazorat 2.13]

Yechim. 0/0 shakli. Ikkala ko'paytuvchiga ajratamiz: $(x^2 + 4x + 3)/(x^2 - 9) = ((x + 1)(x+3))/((x+3)(x-3)) = (x+1)/(x-3), x \neq -3$ bo'lganda. Chekima: $(-3+1)/(-3-3) = (-2)/(-6) = \frac{1}{3}$.

4. $\lim_{x\to 5}(\sqrt{x-1} - 2)/(x - 5)$ **ni hisoblang.**
[Manba: Nazorat 2.14]

Yechim. 0/0 shakli. Konjugat $\sqrt{x-1} + 2$ ga ko'paytiramiz:

$$(\sqrt{x-1} - 2)/(x - 5) = (x - 1 - 4)/((x - 5)(\sqrt{x-1} + 2))$$
$$1/(\sqrt{x-1} + 2)(x \neq 5)$$

demak chekima $1/(\sqrt{5-1} + 2) = \frac{1}{4}$.

5. $\lim_{x\to -3}(1/(x+2)+1)/(x+3)$ **ni hisoblang.**
[Manba: Nazorat 2.15]

Yechim. Murakkab kasrni soddalashtiramiz: $1/(x+2)+1=(x+3)/(x+2)$, shuning uchun ifoda $(x+3)/((x+2)(x+3))=1/(x+2)$ ga teng, $x\neq -3$ bo'lganda. Chekima: $1/(-3+2)=-1$.

6. $\lim \frac{1}{x\to 3x-3}-4$ **ni hisoblang.**

$$1/(x-3)-1/(x+1)$$

[Manba: Nazorat 2.16]

Yechim. x^2- 2x $-3=(x-3)(x+1)$ ni ajratamiz va bir xil kasrga keltiramiz:

$$1/(x-3)-4/((x-3)(x+1))=((x+1)-4)/((x-3)(x+1))$$
$$(x-3)/((x-3)(x+1))=1/(x+1)$$

$$(x\neq 3),$$

demak chekima 1/4.

7. Qismli nazorat: $f(x)=x^2-x,\ x\leq 3$ **bo'lganda va** $f(x)=x+4, x>3$ **bo'lganda; chokdagi bir tomonlama chekimalarni hisoblang.**
[Manba: Nazorat 2.17]

Yechim. Chapda parabola shoxi $\lim_{x\to 3}-f(x)=9$ ni beradi; o'ngda to'g'ri chiziq $\lim_{x\to 3}+(x+4)=7$ ni beradi. Teng emas, demak $\lim_{x\to 3} f(x)$ mavjud emas.

8. $\mathbf{limx}\to 1\frac{x+2}{(x-1)}2$ **ni hisoblang (2.23-misol usuli).**
[Manba: Nazorat 2.18]

Yechim. Surat $\to 3$ (nolga teng emas), maxraj $\to 0^+$ ikkala tomondan ham (kvadrat), shuning uchun nisbat $\to +\infty$: cheksiz chekima, va $x=1$ — vertikal asimptota.

9. Ikki politsiyachi teoremasidan foydalanib, $\lim_{x\to 0} x^2\sin\frac{1}{x}$ **ni hisoblang.**
[Manba: Nazorat 2.19]

Yechim. $-1\leq \sin\frac{1}{x}\leq 1$ ekanligidan, $x^2\geq 0$ ga ko'paytirsak $-x^2\leq x^2\sin 1_x\leq x^2$ hosil bo'ladi. Ikkala baholar ham $\to 0$; (86) bo'yicha chekima 0 ga teng.

10. $\lim \frac{1-cos\theta}{\theta\to 0 sin\theta}$ **ni hisoblang.**
[Manba: Nazorat 2.20]

Yechim. Surat va maxrajni θ ga bo'lamiz (chekima ichida $\theta\neq 0$ bo'lgani uchun qonunli): $1-\cos\theta 1-\cos\theta-\to 0\theta$

$$(1-\cos\theta)/(a\sin\theta)=0,1$$

$\sin\theta\sin\theta\theta$ foydalanib $^{1-cos\theta}\to 0(2.25-\text{misol})va^{sin\theta}\to 1(\text{standart chekima}).\ \theta\theta$

5.3 OpenStax §2.3, mashqlar 83–130 — savollar va to’liq yechimlar

5.3.1 Chekimalar qonunlaridan foydalanish (83–86)

1. $\lim_{x\to 0}(4x^2-2x+3)$**.**
[Manba: §2.3, mashq 83]

Yechim. (85) bo’yicha almashtirish: $4(0)^2-2(0)+3=3$.

2. $\lim \frac{x3+3x2+5}{x\to 1}$ **. 4−7x**
[Manba: §2.3, mashq 84]

Yechim. Maxraj $1da-3$ ga teng $\neq 0$, shuning uchun bo’linma qonuni va almashtirish $\frac{1+3+5}{} = \frac{9}{-3} = -3$ beradi. $-3\surd$

$$\lim_{x\to 9}\sqrt{x^2-6x+3}.$$

3. $\lim_{x\to -2}$
[Manba: §2.3, mashq 85]

Yechim. Ildiz ostidagi ifoda -2 da uzluksiz va musbat; ildiz qonuni: $\sqrt{4+12+3} = \sqrt{19}$.

4. $\lim_{x\to -1}(9x+1)^2$**.**
[Manba: §2.3, mashq 86]

Yechim. Daraja qonuni, keyin almashtirish: $(9(-1)+1)^2=(-8)^2=64$.

5.3.2 To’g’ri almashtirish (87–92)

1. $\lim_{x\to 7} x^2$**.**
[Manba: §2.3, mashq 87]

Yechim. 49.

2. $\lim_{x\to -2}(4x^2-1)$**.**
[Manba: §2.3, mashq 88]

Yechim. $4(4)-1=15$.

3. $\lim \frac{1}{x\to 01+sinx}$**.**
[Manba: §2.3, mashq 89]

Yechim. Maxraj $1+\sin 0=1\neq 0$: bo’linma qonuni $\frac{1}{1}=1$ ni beradi.

4. $\lim_{x\to 2} e^{2x-x2}$**.**
[Manba: §2.3, mashq 90]

Yechim. Ko’rsatkich: $2(2)-4=0$; $e^0=1$.

5. $\lim \frac{2+7x}{x\to 1x+6}$**.**
[Manba: §2.3, mashq 91]

Yechim. $\frac{2+7}{1+6} = 9_7$.

6. $\lim_{x\to 3} \ln e^3 x$**.**
[Manba: §2.3, mashq 92]

Yechim. $\ln e^3 x = 3$x aynan (logarifm va eksponenta o’zaro teskari), shuning uchun limit $3\cdot 3=9$.

5.3.3 Noaniqlik 0/0(93–102)

1. $\lim \frac{x2-16}{x\to 4x-4}$.
[Manba: §2.3, mashq 93]
Yechim. 0/0 shakli. Ajratamiz: $\frac{(x-4)(x+4)}{} = x + 4x \neq 4$ da; limit $= 8$. $x - 4$

2. $\lim \frac{x-2}{x\to 2x2-2x}$.
[Manba: §2.3, mashq 94]
Yechim. Maxrajni ajratamiz: $\frac{x-2}{x(x-2)} = 1_x x \neq 2$ da; limit $= 1_2$.

3. $\lim \frac{3x-18}{x\to 62x-12}$.
[Manba: §2.3, mashq 95]
Yechim.$\frac{3(x-6)}{2(x-6)} = 3_2 x \neq 6$ da; limit $= 3_2$.

$$\lim_{h\to 0}((1+h)^2 - 1)/h$$

4. $\lim_{h\to 0}$. h
[Manba: §2.3, mashq 96]
Yechim.$\frac{1+2h+h2-1}{} = \frac{h(2+h)}{} = 2 + h$; limit $= 2(bux^2$ ning hosilasi $1da)$. hh

5. $\lim \frac{t-9}{t\to 9t-3}$. $\sqrt{}$
[Manba: §2.3, mashq 97]
Yechim. 0/0 shakli. Maxrajning konjugatiga ko'paytiramiz: $\sqrt{}$

$$\lim_{t\to 9}(t-9)/(\sqrt{t}-3) = \lim_{t\to 9}(t-9)(\sqrt{t}+3)/((\sqrt{t}-3)(\sqrt{t}+3))$$
$$\lim_{t\to 9}(t-9)(\sqrt{t}+3)/(t-9) = \lim_{t\to 9}$$

$$(\sqrt{t}+3) = 6(t \neq 9)$$

demak limit $9 + 3 = 6$.

6. $\lim_{h\to 0}(1/(a+h)-1/a)/h$ **ni toping, bu yerda** a **— nol bo'lmagan haqiqiy doimiy.**
[Manba: §2.3, mashq 98]
Yechim. Numeratorni bir xil ko'paytuvchiga keltiramiz: $\frac{a-(a+h)-h-1}{}$

$$\lim_{h\to 0}(a^{a+h} - a^a)/h, buyerdaa—nolbo'lmaganhaqiqiydoimiy.$$

Limit: $-\frac{1}{a2}(1/x$ ning hosilasi $ada)$.

7. $\lim \frac{sin\theta}{\theta\to\pi tan\theta}$.
[Manba: §2.3, mashq 99]
Yechim. $\theta \neq \pi$ yaqinida π da: $\frac{sin\theta}{tan\theta} = \sin\theta \cdot \cos\frac{\theta}{sin\theta} = \cos\theta$ (qisqartish qonuniy, chunki u yerda $\sin\theta \neq 0$). Limit: $\cos\pi = -1$.

8. $\lim \frac{x3-1}{x\to 1x2-1}$.
[Manba: §2.3, mashq 100]

Yechim. Ikkalasini ham ajratamiz: $^{(x-1)(x2+x+1)}$ $=^{x2+x+1}$; limit $= \frac{3}{2}$.

$$(x-1)(x+1)/(x+1)$$

9. $\lim \frac{2x2+3x-2}{x\to 1/2}$ **. 2x−1**

[Manba: §2.3, mashq 101]

Yechim. $2x^2+ 3x -2 = (2x-1)(x+2)$; $x+2$ gacha qisqartiramiz; limit $= \frac{1}{2} + 2 = 5_2$.

$$\sqrt{x+4} - 1$$

10. $\lim_{x\to -3}$**.** $x+3$

$$\sqrt{x+4+1}$$

[Manba: §2.3, mashq 102]

Yechim. Konjugat: $\sqrt{x+4+1}$ ga ko'paytiramiz:

$$((x+4)-1)/\sqrt{x+4+1} = 1/((x+3)(x+4+1)), (x \neq -3);$$

limit $= \frac{1}{1+1} = 1_2$.

5.3.4 Shakllar $K/0$: bir tomonli cheksiz limitlar (103–106)

1. $\lim_{x\to -2} {}^{-}{}^{2x2+7x-4}$

$$(x+2)/(x-2)$$

[Manba: §2.3, mashq 103]

Yechim. Ajratamiz: numerator $= (2x-1)(x+4)$, maxraj $= (x+2)(x-1)$. $x=-2$ da numerator $(-5)(2) = -10 \neq 0$, maxraj esa $\to 0$: $K/0$ shakli. -2 yaqinidagi ishora: $x = -2.1$ ni olaylik: $(x+2) < 0, (x-1) < 0$, demak maxraj musbat; numerator taxminan $-10 < 0$. Nisbat $\to -\infty$.

2. $\lim_{x\to -2} {}^{+}{}^{2x2+7x-4}$

$$(x+2)/(x-2)$$

[Manba: §2.3, mashq 104]

Yechim. Xuddi shu numerator; endi $(x+2) > 0, (x-1) < 0$: maxraj manfiy. Nisbat $\to +\infty$.

3. $\lim_{x\to 1} {}^{-}{}^{2x2+7x-4}$

$$(x+2)/(x-2)$$

[Manba: §2.3, mashq 105]

Yechim. 1 da numerator: $(1)(5) = 5 > 0$. $x \to 1^-$ da: $(x+2) > 0, (x-1) < 0$, maxraj manfiy: nisbat $\to -\infty$.

4. $\lim_{x\to 1} {}^{+}{}^{2x2+7x-4}$

$$(x+2)/(x-2)$$

[Manba: §2.3, mashq 106]

Yechim. Maxraj 1 dan o‘ngda musbat: nisbat $\to +\infty$. (Ikkala tomonlama chegaralar $-2va1$ da mavjud emas; $x=-2vax=1$ — vertikal asimptotlar.)

5.3.5 Berilgan chegaralar bilan chekalar qonunlari (107–114)

Butun davomida: $\lim_{x\to 6} f(x)=4$, $\lim_{x\to 6} g(x)=9$, $\lim_{x\to 6} h(x)=6$.

1. $\lim_{x\to 6}$ **2f** $(x)g(x)$**.**
[Manba: §2.3, mashq 107]

Yechim. Ko‘paytma, keyin doimiy ko‘paytuvchi: $2\cdot 4\cdot 9=72$.

$$: g(x)-1$$

2. $\lim_{x\to 6} f(x)$**}.**
[Manba: §2.3, mashq 108]

Yechim. Bo‘linma qonuni: $^{9-1}=2$.

$$\lim_{x\to 6} f(x).$$

3. $\lim_{x\to 6} f(x)+\frac{1}{3}g(x)$**.**
[Manba: §2.3, mashq 109]

Yechim. Yig‘indi va doimiy ko‘paytuvchi: $4+\frac{9}{3}=7$.

4. $\lim_{x\to 6} h(x)$
[Manba: §2.3, mashq 110]

Yechim. Daraja qonuni: $6^3=216$. p

5. $\lim_{x\to 6} g(x)-f(x)$**.**
[Manba: §2.3, mashq 111]

Yechim. Ayirma, keyin ildiz: $\sqrt{9}-4=\sqrt{5}$.

6. $\lim_{x\to 6} xh(x)$**.**
[Manba: §2.3, mashq 112]

Yechim. Ko‘paytma qonuni $\lim x=6$ bilan: $6\cdot 6=36$.

7. $\lim_{x\to 6}[(x+1)f(x)]$**.**
[Manba: §2.3, mashq 113]

Yechim. $(6+1)\cdot 4=28$.

8. $\lim_{x\to 6} f(x)g(x)-h(x)$**.**
[Manba: §2.3, mashq 114]

Yechim. $4\cdot 9-6=30$.

5.3.6 Qismlarga bo‘lingan funksiyalar va grafiklar bo‘yicha chegaralar (115–125)

(

$$x^2 \le 3$$

1. $f(x) = x > 3$**:** a **ni toping.** $\lim_{x\to 3} -f, b$**.** $\lim_{x\to 3} +f$**.** $x + 4$
[Manba: §2.3, mashq 115]

Yechim. a. Parabola shoxi: 9. b. Chiziqli shox: 7. (Ikkala tomonlama chek mavjud emAS.) (

$$(x^3 - 1)/x \le 0$$

2. $g(x) = x > 0$**:** a **ni toping.** $\lim_{x\to 0} -g, b$**.** $\lim_{x\to 0} +g$**. 1**

$$0^3 - 1 = -1.$$

[Manba: §2.3, mashq 116]

Yechim. a. Kubik shox: b. Doimiy shox: 1. (Ikkala tomonlama chek mavjud emAS.) (

$$x^2 - 2x + 1, x < 2$$

3. $h(x) = x \ge 2$**:** a **ni toping.** $\lim_{x\to 2} -h, b$**.** $\lim_{x\to 2} +h$**.** $3 - x$
[Manba: §2.3, mashq 117]

Yechim. a. $4 - 4 + 1 = 1$. b. $3 - 2 = 1$. Ikkala tomon ham mos keladi: $\lim_{x\to 2} h(x) = 1 = h(2)$ — u yerda uzluksiz, qismlarga bo‘lingan ta’rifiga qaramay.

4. $fvag$ **ikkita tasvirlangan funksiya bo‘lsin (V shaklidagi** $f(x) = |x + 2|$**; chiziqli chap shoxi** $g(x) = -x - 6$ **va egri chiziqli o‘ng shoxi bo‘lgan qismlarga bo‘lingan** g**),** $\lim_{x\to -3} +(f(x) + g(x))$ **ni hisoblang.**
[Manba: §2.3, mashq 118]

Yechim. Grafikdan: $f(-3) = |-3+2| = 1$ va egri chiziqli shoxda $g(-3^+) = 1$. Yig‘indi qonuni: $1 + 1 = 2$.

$$f(-3) - 3g(-3^-) = 1 - 3(-1) =$$

[Manba: §2.3, mashq 119]

Yechim. Chapda chiziqli shox $g(-3^-) = -(-3) - 6 = -3...$ grafikdagi ulanishni o‘qib, -3 dagi chap shoxning qiymati -1; grafik-qiymatni olib $g(-3^-) = -1$: $f(-3)-$ 3g$(-3^-) = 1 - 3(-1) = 4$.

6. $\lim_{x\to 0}$**. 3**
[Manba: §2.3, mashq 120]

Yechim. $f(0) = |2| = 2$; $g(0) = -2$ (egri chiziq (0, -2) dan o‘tadi). Ko‘paytma va bo‘linma: $^{2\cdot(-2)} = -\frac{4}{3}$. 3

$$7. \lim_{x\to 3}(2 + g(x))$$

7. $\lim_{x\to -3} f(x)$**}.**

$$(2 + g)/(-3 - 2) + (-1)/(2 + 1)$$

[Manba: §2.3, mashq 121]

Yechim. Chapda: $\lim_{x\to -3} f(x)\} = 1$; o‘ngda: $\lim_{x\to -3} f(x)\} = 3$. Bir tomonlama

$$f(-3) = \frac{1}{2}$$

cheklar farq qiladi, demak ikkala tomonlama chek mavjud emas. $_2$.

8. $\lim_{x\to 1} f(x)$

[Manba: §2.3, mashq 122]

Yechim. $f(1) = |3| = 3$; daraja qonini: 9.

$$P_3 = (1)/(3); kanonikshakl : 3.$$

9. $\lim_{x\to 1} f(x) - g(x)$**.**

[Manba: §2.3, mashq 123]

Yechim. Grafikdan, $g(1) = -5 va f(1) = 3$: ayirma 8; kub ildiz: $_3\sqrt{8} = 2$.

10. $\lim_{x\to -7} xg(x)$**.**

[Manba: §2.3, mashq 124]

Yechim. $x = -7$ da chiziqli shox $g(-7) = -(-7) - 6 = 1$ beradi; ko‘paytma: $(-7)(1) = -7$.

11. $\lim_{x\to -9}[xf(x) + 2g(x)]$**.**

[Manba: §2.3, mashq 125]

Yechim. $f(-9) = |-7| = 7$; $g(-9) = 9 - 6 = 3$; $(-9)(7) + 2(3) = -63 + 6 = -57$.

5.3.7 Ikki politsiyachi teoremasi va uning qo‘llanilishi (126–130)

T To‘g‘ri yoki noto‘g‘ri: agar 2x$-1 \le g(x) \le x^2-$2x$+3x = 2$ yaqinida bo‘lsa, u holda $\lim_{x\to 2} g(x) = 0$.

[Manba: §2.3, mashq 126]

Yechim. Noto’g’ri. Ikkala baham ham $x = 2da 2(2) - 1 = 3 = 4 - 4 + 3$ ga teng, shuning uchun ”ikki militsioner” teoremasiga ko’ra $\lim_{x\to 2} g(x) = 3$, nol emas. (Teorema baholarning umumiy limitini aniq beradi.)$T \lim_{\theta\to 0} \theta^2 \cos\frac{1}{\theta}$.

[Manba: §2.3, mashq 127]

Yechim. $-1 \le \cos\frac{1}{\theta} \le 1$ beradi $-\theta^2 \le \theta^2 \cos 1_\theta \le \theta^2$; ikkala baham ham $\to 0$; siqilish: limit $= 0$. (0, x ratsional)

1. $\lim_{x\to 0} f(x)$**, bunda** $f(x) = x$ **irratsional.**

$$\lim_{x\to 0} f(x) = 0$$

[Manba: §2.3, mashq 128]

Yechim. Barcha x uchun: $0 \le f(x) \le x^2$ (ratsional $x0$ beradi; irratsional $xx^2 \ge 0$ beradi). $0 \to 0 va x^2 \to 0$ bo’lgani uchun, ikki politsiyachi teoremasi $\lim_{x\to 0} f(x) = 0$ ni majburlaydi — hatto $f harx \ne 0$ nuqtada g’alati tarzda uzilgan bo’lsa ham. $^q 10^{-19}$; $\lim_{r\to 0+} E(r) va T$ ni hisoblang Kulon qonuni: $E(r) = q =$

$$E(r) = fracq4\pi\varepsilon_0 r^2$$

tushuntiring; nima uchun o'ng tomonda?

[Manba: §2.3, mashq 129]

Yechim. Surat — musbat doimiy; $r \to 0^+$ maxraj $\to 0^+$, demak $E(r) \to +\infty$: maydon kattaligi cheksiz oshadi, sinov nuqtasi zaryadga yaqinlashganda. Faqat $r \to 0^+$ ma'noga ega, chunki r — masofa va manfiy bo'lishi mumkin emas ($var = 0$—bu zaryadning o'zi joylashgan joyi). ρ zichligi $= m/VdanV(\rho) = \frac{m}{\rho}$; $m = 8$ kg bo'lganda $\lim_{\rho\to 0} + V(\rho)$ ni hisoblang va fizik ma'nosini tushuntiring.

[Manba: §2.3, mashq $130]_\rho$

; $\rho \to 0^+$ hajm $\to +\infty$:

Yechim. $V(\rho) =$ massasi doimiy, zichligi esa yo'qolayotgan jism cheksiz hajmni egallashi kerak («ideal gazning noziklashuvi» degan rasmiy bayonot). O'ng tomondan yaqinlashish — bu yagona fizik variant, chunki zichlik musbat.

5.4 4-hafta konspekti yoritilgan misollar — savollar va to'liq yechimlar

1. **Isbotlang, ki** $\lim_{x\to 1}(x^2-1)/(x-1)\frac{0}{0}$ **shakliga ega, va uni hisoblang;** $\lim_{x\to 0} x/x$, $\lim_{x\to 0} x^2/x$, $\lim_{x\to 0} x/x^2$ **bilan solishtiring.**

   [Manba: 4-hafta konsperti, noaniqliklar]

   Yechim. $x = 1$ qo'yish 0/0 — noaniqlik beradi. Yoyib: $(x-1)(x+1) = x+1, x \neq 1$ bo'lganda, demak limit 2. Uchta solishtirish $x-1$

   $$\lim_{x\to 1}(x^2-1)/(x-1), \lim_{x\to 0} x^x, \lim_{x\to 0} x^2, \lim_{x\to 0} x^x$$

   2

   bir xil 0/0 shaklini uchta turli natijaga olib keladi: $= 1 \to 1$; $= x \to 0$; $=$

   $$\lim_{x\to 1}(x^2-1)/(x-1)$$

   $x^1 \to \pm\infty$ (ikkala tomondan chekli limit mavjud emas). Shuning uchun 0/0 avtomatik qiymat bermaydi: natija surat va maxrajning nolga qanday yaqinlashishiga bog'liq.

2. **Standart limit** $\lim_{x\to 0}$ **sinx**$/x = 1$ **dan foydalanib,** $\lim_{x\to 0} \sin^5 x/x$ **ni hisoblang.**

   [Manba: 4-hafta konsperti, standart limitlar]

   Yechim. $u =$ 5x almashtirish ($x \to 0$ bo'lganda $u \to 0$): sin 5x$/x = 5 \cdot$ sin 5x$/(5x) =$ 5$\cdot$sinu$/u \to 5 \cdot 1 = 5$.

3. **Asimptotik ekvivalentlikdan foydalanib,** $\lim_{x\to 0} \sin$ **3x/5x va** $\lim_{x\to 0}(1-\cos x)/x$ **ni hisoblang.**

   $$\sin^3 x \sim 3x/5$$

   [Manba: 4-hafta konsperti, asimptotika]

   Yechim. $x \to 0$ bo'lganda sin 3x ~ 3x bo'lgani uchun, sin 3x/5x ~ 3x/5x $= \frac{3}{5}$, demak limit 3/5. $1 - \cos x \sim x^2/2$ bo'lgani uchun, $(1-\cos x)/x \sim x^2/2$x $= x/2 \to 0$, demak limit 0.

   $$\lim_{x\to 0}(\sin 3x)/(5x) = \frac{3}{5}$$

standart limit (87) ekvivalentlik orqali takrorlanadi.

4. Isbotlang, ki $3x^2+5x-1 \sim 3x^2 x \to \infty da$, va $\lim_{x\to\infty}(2x^3+x)/(5x^2-1)$ ni ekvivalentliklar orqali hisoblang.

[Manba: 4-hafta konsperti, cheksizdagi ekvivalentlik]

Yechim. $(3x^2+5x-1)/(3x^2) = 1+5/(3x)-1/(3x^2) \to 1$, $x \to \infty da$, shuning uchun $3x^2+5x-1 \sim 3x^2$.

$$\lim_{x\to\infty}(3x^2+5x-1)/(3x^2) = 1+0-0 = 1, ekvivalentliko'rnatildi:$$

yuqori darajali had uzoq muddatli xulq-atvorni boshqaradi. U holda $(2x^3+x)/(5x^2-1) \approx (\frac{2}{5})x$.

$$\lim_{x\to\infty}(x^3+x)/(3x^2-1)$$

$= \frac{2}{5}x$, bu $+\infty$ ga intiladi; aniqrog'i, asl kasrning $\frac{2}{5}x$ nisbati 1 ga intiladi.

5. Isbotlang, ki $\sin x = O(x) x \to 0 da$, va $3x = O(x)$, lekin $3x \neq o(x)$.

[Manba: 4-hafta konspekti, Landau belgilari]

Yechim. $|\sin x| \le |x|$ tengsizligi barcha haqiqiy x lar uchun to'g'ri (standart geometrik fakt), bu esa aynan

$$|\sin x| \le 3|x|$$

cheklanganlikni bildiradi $_{|x|}$: $\sin x = O(x)$. 3x: $= 3$ cheklangan, demak $|x|$ 3x $= O(x)$; lekin $\frac{3x}{x} = 3 \to 0$, shuning uchun $3x \neq o(x)$. $BuOvao$ o'rtasidagi farqning eng ravshan misolidir: katta O har qanday cheklangan nisatni qabul qiladi, kichik o esa nisatning nolga intilishini talab qiladi.

6. $x^2 = o(x) nix \to 0 davax = o(x^2)$, $x^{10} = o(e^x) nix \to \infty$ da tekshiring.

$$x^2 - 2x + x = o(x) nolgayaqin.$$

[Manba: 4-hafta konspekti, Landau belgilari]

Yechim. $= x \to 0$: Cheksizlikda $=$

$$x^2 = o(x) x \to 0 da. (16)$$

$o(x^{1/2})$; $x^{10}/e^x \to 0$ standart limit $\lim_{x\to\infty} x^n/e^x \to 0 (88)$ bo'yicha: eksponensial o'sish oxir-oqibat har qanday chekli darajadan ustun keladi.

7. $\sin x \sim xvae^x = 1+x+\frac{x2}{2}+o(x^2)$ ni kichik o tili bilan yozing. Nima uchun $f(x) = O(g(x))$ oddiy tenglama emas?

[Manba: 4-hafta konspekti, nisbatlar]

Yechim. (94) ga ko'ra, $f \sim g$ bo'lishi $f = g + o(g)$ bo'lgandagina to'g'ri: demak, $\sin x \sim xni \sin x = x + o(x)$ deb yozish mumkin — bu x ga nisbatan e'tiborsizlik qilingan xato. Eksponentiya uchun $e^x = 1 + x + o(x)$ to'g'ri, chunki $^{ex-1-x} \to 0, x$ va yana bir hadni saqlab qo'yish $e^x = 1 + x + \frac{x2}{2} + o(x^2)$ ni beradi. Nihoyat, $f = O(g)$ tenglama emas, chunki $O(g)g$ ga doimiy ko'paytmalar bilan chegaralangan butun funksiyalar sinfini bildiradi; $2x^2+3x+1 = 2x^2 + O(x)$ yozuvi $3x+1$ qoldig'i x ning doimiy ko'paytmasidan tezroq o'smasligini anglatadi.

6 Darsliklardan qo’shimchalar — to’liq yechimli majburiy o’qish uchun qo’shimcha mashqlar

6.1 OpenStax §1.2 (Asosiy funksiyalar sinflari) — tanlangan mashqlar yechilgan

1. Har bir nuqta juftligi uchun ular orqali o’tgan to’g’ri chiziqning burchak koeffitsiyentini toping va aytoring, to’g’ri chiziq o’sayotgan, kamayayotgan, gorizontal yoki vertikal bo’ladimi: $(59)(-2,\ 4)va(1,\ 1)$**;** $(61)(3,\ 5)va(-1,\ 2)$**;** $(63)(2,\ 3)va(5,\ 7)$**;** $(65)(2,\ 4)va(1,\ 4)$**;** $(66)(1,\ 4)va(1,\ 0)$**.**

[Manba: OpenStax §1.2, masala 59–66]

Yechim. Burchak koeffitsiyenti $m = (y_2 - y_1)/(x_2 - x_1)$. $(59)m = (-5-1)/(2-(-4)) = -\frac{6}{6} = -1$: kamayayotgan. $(61)m = 4 > 0$: o’sayotgan. $(63)m = (7-3)/(5-2) = \frac{4}{3}$: o’sayotgan. $(65)m = (4-4)/(1-2) = 0$: gorizontal. $(66)\Delta x = 0$: vertikal, koeffitsiyent aniqlanmagan.

2. Tenglamani qiyalik-siljish shaklida yozing: (67) qiyalik −6, (1, 3) nuqtadan o’tuvchi; (70) qiyalik $\frac{2}{5}, x$ **o’qi bilan kesishish nuqtasi 8;** $(71)(2,\ 1)va(-2,\ -1)$ **nuqtalardan o’tuvchi; (73) kesishish nuqtalari:** $xda5vayda-3$**.**

[Manba: OpenStax §1.2, masala 67–73]

Yechim. $(67)y - 3 = -6(x-1) \Rightarrow y = -6x + 9$. $(70)x$ o’qi bilan kesishish nuqtasi 8 degani (8, 0) nuqta: $y - 0 = \frac{2}{5}(x-8) \Rightarrow y = 2_5x - \frac{16}{5}$. $(71)m = -\frac{1-1}{-2-2} = -\frac{2}{-4} = 1_2$; $y - 1 = \frac{1}{2}(x-2) \Rightarrow y = 1_2x$. $(73)(5,0)va(0,-3)$ nuqtalardan o’tuvchi: $m = -\frac{3-0}{0-5} = 3_5$; $y = 3_5x - 3$.

3. (75–76) Har bir chiziqli tenglama uchun qiyalik m **va siljish** b **ni ko’rsating:** $y = \mathbf{2x - 3}$**;** $y = -\frac{1}{7}x + 1$**.**

[Manba: OpenStax §1.2, masala 75–76]

Yechim. $(75)m = 2, b = -3$. $(76)m = -\frac{1}{7}, b = 1$.

4. (81–82) Qiyalik va siljish shaklida qayta yozing: $\mathbf{2x} + \mathbf{3y} = 6$**;** $\mathbf{6x - 5y} + 15 = 0$**.**

[Manba: OpenStax §1.2, masala 81–82]

Yechim. $(81)y = 2 - \frac{2}{3}x$: $m = -\frac{2}{3}, b = 2$. $(82)y = 6_5x + 3$: $m = 6_5, b = 3$.

5. (83–86) Har bir ko’phad uchun: daraja; nollari; y **bo’yicha siljish; cheklarda xulq-atvori; juftligi.** a**.** $f(x) = \mathbf{2x^2 - 3x - 5}$**;** c**.** $f(x) = x^3 + \mathbf{3x^2} - x - 3$**.**

[Manba: OpenStax §1.2, masala 83–86]

Yechim. a. Daraja 2; $2x^2 - 3x - 5 = (2x-5)(x+1)$: nollari $\frac{5}{2}va - 1$; siljish −5; burchaklar yuqoriga (bosh koeffitsient $2 > 0$), demak $f \to +\infty$ ikkala chekkada ham; juft ham toq emas $(toqhad - 3xva$ doimiy had tufayli). c. Daraja 3; $x = -1$ sinab ko’ramiz: $-1 + 3 + 1 - 3 = 0$, demak $(x+1)$ — ko’paytuvchi: $x^3 + 3x^2 - x - 3 = (x+1)(x^2-1) = (x+1)(x-1)(x+1) = (x-1)(x+1)^2$: nollari 1 (oddiy)$va - 1$(ikkita); siljish −3; kub bosh koeffitsienti musbat: $f \to +\infty x \to +\infty da, f \to -\infty x \to -\infty$ da; juft ham toq emas.

6. $(88-91)g$ **grafigini asosiy grafikdan foydalanib tasviring:** $(88)g(x) = x^2 - 1$**;** $(89)g(x) = (x+3)^2 + 1$**;** $(90)g(x) = \sqrt{x} + 2$**;** $(91)g(x) = -\sqrt{x} - 1$**.**

[Manba: OpenStax §1.2, mashq 88–91]

Yechim. $(88)y = x^2$ parabolasining pastga 1 birlik siljishi. (89) chapga 3 va yuqoriga 1 birlik siljigan parabola (cho’qqisi $(-3,\ 1)$). (90) ildiz grafigining

chapga 2 birlik siljishi (aniqlanish sohasi $[-2,\infty)$). (91)ildiz grafigining x o'qi bo'yicha aks ettirilishi va pastga 1 birlik siljishi (aniqlanish sohasi $[0,\infty)$, qiymatlar ≤ -1). (

$$f(x) = 4x + 3, x \le 0; (9)$$

7. $(94) f(x) = x > 0$**:** $f(-3), f(0), f(2)$ **ni hisoblang; chizing.**

$$-x + 1$$

[Manba: OpenStax §1.2, mashq 94]

Yechim. $f(-3) = 4(-3) + 3 = -9$ (chap shox); $f(0) = 3$; $f(2) = -2 + 1 = -1$ (o'ng shox). Grafik: (0, 3) gacha o'sayotgan to'g'ri chiziq, keyin ochiq nuqtadan (0, 1) boshlanayotgan kamayayotgan to'g'ri chiziq — 0 da sakchikli uzilish. (

$$f(x) = (x-3)/(x-2), x \ne 2$$

8. $(97) g(x) = x = 2$**:** $g(0), g(-4), g(2)$ **ni hisoblang. 4,**

[Manba: OpenStax §1.2, mashq 97]

Yechim. $g(0) = \frac{3}{-2} = -\frac{3}{2}$; $g(-4) = \frac{3}{-6} = -\frac{1}{2}$; $g(2) = 4$ maxsus ta'rifga ko'ra (formula aniqlanmagan — shuning uchun qismiy qiymat berilgan). $^3\sqrt{x}$ —

9. To'g'ri yoki noto'g'ri (98–103): $(98) f(x) = \frac{4x+1}{7x-2}$ **transsendent;** $(99) g(x) =$ **toq ildiz funksiyasi; (100) logarifmik funksiya algebraik;** $(101) f(x) = x^b$ **haqiqiy konstanta** b **bilan — eksponensial funksiya; (102) toq darajali ildizning aniqlanish sohasi — barcha haqiqiy sonlar; (103) kompaniya $20,500 evaziga uskunani sotib oldi, 3 yildan keyin $12,300 ga teng —** $V(t)$ **ni toping va tushuntiring.**

[Manba: OpenStax §1.2, mashq 98–103]

Yechim. (98) Noto'g'ri: ko'phadlar nisbati ratsional, demak algebraik, transsendent emas. (99) To'g'ri: kub ildizlari barcha haqiqiy sonlar uchun aniqlangan va funksiya toq. (100) Noto'g'ri: x^b — daraja funksiyasi; logarifmlar transsendent. (101) Noto'g'ri: eksponentning o'zgaruvchisi darajada, b^x. (102) Noto'g'ri: toq darajali ildizlar manfiy bo'lmagan radikaldan talab qiladi, shuning uchun aniqlanish sohasi manfiy sonlarni istisno qiladi (toq ildizlar barcha haqiqiy sonlarni qabul qiladi). (103) Chiziqli amortizatsiya: $V(t) = 20500 + ^{12300-20500}\, t = 320500 - \frac{8200}{3}t \approx 20500 - 2733.33t$. $V(0) = 20500$ — sotib olish narxi; t bilan kesishish $t \approx 7.5$ yilda — qiymatning nolga aylanish vaqti; $V(5) = 20500 - \approx$ \$6833. 3^{41000}

6.2 OpenStax §1.5 (Eksponensial va logarifmik funksiyalar) — tanlangan mashqlar yechimi

1. (239–245) Chizing; aniqlanish sohasini, qiymatlar to'plamini, gorizontal asimptotani ko'rsating: *a.* $f(x) = e^x + 2$**;** *b.* $f(x) = -2^x$**;** *c.* $f(x) = 1 - 2^{-x}$**;** *d.* $f(x) = e^{-x} - 1$**.**

[Manba: OpenStax §1.5, mashq 239–245]

Yechim. *a.* e^x, yuqoriga 2 birlik siljishi: aniqlanish sohasi R, qiymatlar $(2,\infty)$, asimptota $y = 2$. *b.* x o'qi bo'yicha aks ettirish: aniqlanish sohasi R, qiymatlar $(-\infty, 0)$, asimptota $y = 0$. *c.* $x \to +\infty da, 2^{-x} \to 0$, demak $f \to 1$ pastdan;

$x \to -\infty da, 2^{-x} \to \infty$, demak $f \to -\infty$: o'sadi, qiymatlar $(-\infty, 1)$, asimptota $y = 1$. *d.* e^x aks ettirilgisi (o'ngga zaiflashadi): aniqlanish sohasi R, qiymatlar $(-1, \infty)$, asimptota $y = -1$.

2. (246–253) Ekvivalent eksponensial shaklda qayta yozing: *a.* $\log_3 81 = 4$**;** *b.* $\log_8 2 = \frac{1}{3}$**;** *c.* $\log_5 1 = 0$**;** *d.* $\ln 1_{e3} = -3$**.**

[Manba: OpenStax §1.5, mashq 246–253]

Yechim. (39) ga ko'ra: *a.* $3^4 = 81$. *b.* $8^{1/3} = 2$. *c.* $5^0 = 1$. *d.* $e^{-3} = 1/e^3$.

3. (254–262) Ekvivalent logarifmik shaklda qayta yozing: *a.* $2^3 = 8$**;**

b. $4^{-2} = \frac{1}{16}$**;** *c.* $10^2 = 100$**;** *d.* $3^4 = 81$**;** *e.* $3^3 = 27$**;** *f.* $9^y = 150$**.**

[Manba: OpenStax §1.5, mashq 254–262]

Yechim. *a.* $\log_2 8 = 3$. *b.* $\log_{1/4} 16 = -2$. *c.* $\log 100 = 2$. *d.* $\log_{1/3} x$ ko'rsatkichni anglatadi: $(\frac{1}{3})^y = x$; $\log_{64} 4 = \frac{1}{3}$. *f.* $\log_9 150 = y$.

4. (264–266) Chizing; aniqlanish sohasini, qiymatlar to'plamini, vertikal asimptotani ko'rsating: *a.* $f(x) = 3 + \ln x$**;** *b.* $f(x) = \ln(x - 1)$**;** *c.* $f(x) = \ln(-x)$**.**

[Manba: OpenStax §1.5, mashq 264–266]

Yechim. *a.* $\ln x$, yuqoriga 3 birlik siljishi: aniqlanish sohasi $(0, \infty)$, qiymatlar R, vertikal asimptota $x = 0$. *b.* o'ngga 1 birlik siljishi: aniqlanish sohasi $(1, \infty)$, qiymatlar R, asimptota $x = 1$. *c.* y o'qi bo'yicha aks ettirish: aniqlanish sohasi $(-\infty, 0)$, qiymatlar R, asimptota $x = 0$.

5. (270–275) Logarifm qonunlari bo'yicha oching: *a.* $\log x^4$**;** *b.* $\log \frac{9a3}{b}$**;** *c.* $\ln ab$**;** *d.* $p \log_5$ **125xy^3.**

[Manba: OpenStax §1.5, mashq 270–275]

Yechim.$_3$*a.* $4 \log x$. *b.* $\log 9 + 3 \log a - \log b = 2 \log 3 + 3 \log a - \log b$. *c.* $\ln a + \ln b$. *d.* $\frac{11}{5} \log_5 (125\text{xy}^3) = 1_5(3 + \log_5 x + 3 \log_5 y)$.

6. (276–283) Aniq yeching: *a.* $5^x = 125$**;** *b.* $e^3 x - 15 = 0$**;** *c.* $8^x = 4$**;** *d.* $4^{x+1} - 32 = 0$**;** *e.* $3^{x/14} = \frac{1}{10}$**;** *f.* $10^x = 7.21$**.**

[Manba: OpenStax §1.5, mashq 276–283]

Yechim. *a.* $x = 3$ (chunki $125 = 5^3$). *b.* $3\text{x} = \ln 15, x = \frac{ln15}{} \approx 0.9027$. *c.* $2^3 x = 2^2, 3x = \frac{2}{3}$. *d.* $4^{x+1} = 2^5 \Rightarrow 2^{2x+2} = 2^5 \Rightarrow x = 3\frac{x}{14} \ln 3 = -\ln 10$, $x = -\frac{14ln10}{2}$. *e.* ≈ -29.32. *f.* $\ln 3x = \log 7.21 \approx 0.8580$.

7. (284–291) Agar mumkin bo'lsa, aniq yeching: *a.* $\log_3 x = 0$**;** *b.* $\log_5 x = -2$**;** *c.* $\log_3(x + 5) = 0$**;** *d.* $\log_2(2x - 7) = 0$**;** *e.* $\ln \sqrt{x} + 3 = 2$**;** *f.* $\log_6(x + 9) + \log_6 x = 2$**;** *g.* $\log_4(x + 2) - \log_4(x - 1) = 0$**.**

[Manba: OpenStax §1.5, mashq 284–291]

Yechim. *a.* $x = 3^0 = 1$. *b.* $x = 5^{-2} = \frac{1}{25}$. *c.* $x + 5 = 3^0 = 1, x = -4$; aniqlanish sohasini tekshiramiz $x+5 > 0$: $1 > 0 OK$, demak $x = -4$ to'g'ri ($\log_3 1 = 0$). *d.* $2\text{x} - 7 = 1, x = 4$. *e.* $\sqrt{x} + 3 = e^2$, $x = e^4 - 3 \approx 51.60$. *f.* Birlashtiramiz:

$$\log_6(x(x + 9)) = 2$$

$x(x + 9) = 36 \Rightarrow x^2 + 9\text{x} - 36 = 0 \Rightarrow (x + 12)(x - 3) = 0$; aniqlanish sohasi $x > 0$ talab qiladi, demak $x = 3$. *g.* $\log \frac{x+2}{4x-1} = 0 \Rightarrow \frac{x+2}{x-1} = 1 \Rightarrow x + 2 = x - 1$: imkonsiz; yechim yo'q.

8. (292–297) Asosni o'zgartirish, to'rtta belgigacha yaqinlashtirish: *a.* $\log_5 47$**;** *b.* $\log_2 82$**;** *c.* $\log_{0.5} 211$**;** *d.* $\log_{0.2} 0.452$**.**

[Manba: OpenStax §1.5, mashq 292–297]

Yechim. (41) bo‘yicha: $a.$ $\frac{ln47}{ln5} \approx 2.3922$. $b.$ $\frac{ln82}{ln2} \approx 6.3576$. $c.$ $\frac{ln211}{ln0.5} \approx -7.7181$. $d.$ $\frac{ln0.452}{ln0.2} \approx 0.3536$.

9. (307) Bakteriya koloniyasi 12 soatda ikki barobar ko‘payadi, boshlang‘ich 1000. $a.$ $Q = Q_0 e^{kt}$ dan foydalanib k ni toping; $b.$ 200,000 gacha qancha vaqt kerak?

[Manba: OpenStax §1.5, mashq 307]

Yechim. $a.$ Ikkilash: $2 = e^{12}k$, demak $k = \frac{ln2}{12} \approx 0.0578$ soatda. $b.$ $200000 = 1000\mathrm{e}^{kt}$: $e^{kt} = 200$, $t = \frac{ln200}{} = \frac{12ln200}{} \approx 92.1$ soat. $k \ln 2$

10. (308) Quyonlar populyatsiyasi har 6 oyda ikki barobar ko‘payadi, boshlang‘ich 120. $a.$ $P = P_0 a^t$ dan foydalanib a ni toping; $b.$ 3500 ga qachon yetadi?

[Manba: OpenStax §1.5, mashq 308]

Yechim. $a.$ $a = 6$ oylik davrda (yiliga: t yillarda bo‘lsa $a = 4$; 6 oylik $a = 2$). $b.$ $3500 = 120 \cdot 2^t$(t6 oylik birliklarda): $2^t = \frac{175}{6} \approx 29.17$, $t = \log_2 29.17 = \frac{ln29.17}{} \approx 4.87$ davr, ya’ni taxminan 2.4 yil. $\ln 2$

11. (309) 1906-yilgi San-Fransisko zilzilasi 8.3 magnitudaga ega edi; xuddi shu yapon zilzilasi 4.9. San-Fransisko zilzilasi qanchalik kuchli edi (Rixter shkalasi bo‘yicha)?

[Manba: OpenStax §1.5, mashq 309]

Yechim. Rixter bo‘yicha kuchlilik nisbati: $10^{8.3-4.9} = 10^{3.4} \approx 2512$: San-Fransisko zilzilasi taxminan 2500 barobar kuchliroq — (40) dan logarithmik shkalaning namoyishi.

6.3 OpenStax §2.2 — jadval mashqlari (42–45) yechilgan

1. Jadval bo‘yicha $\lim_{t\to 0+} \cos t/t$ ni baholang $(t = 0.1,\ 0.01,\ 0.001,\ 0.0001)$.

[Manba: OpenStax §2.2, mashq 42]

Yechim. Qiymatlar: $t = 0.1$: $0.\frac{995004}{0}.1 = 9.95004$; $t = 0.01$: $0.\frac{99995}{0}.01 = 99.995$; $t = 0.001$: $0.9999995/0.001 = 999.9995$; $t = 0.0001$: 9999.95. Qiymatlar cheksiz o‘sadi: surat $\to 1$, maxraj $\to 0^+$ — $K/0^+$ shakli, shuning uchun limit $+\infty$.

$$\lim_{t\to 0+} \cos(t)/t = +\infty$$

2. Jadval bo‘yicha $\lim_{x\to 2}(x-2)/(x^3-4x)$ ni baholang.

[Manba: OpenStax §2.2, mashq 43]

Yechim. Avval soddalashtiramiz: $(x-2)/(x^3-4x) = (x-2)/(x(x-2)(x+2)) = 1/(x(x+2))$, $x \neq 2$ bo‘lganda. Jadval qiymatlari: $x = 1.99$: $1/(1.99 \cdot 3.99) = \frac{1}{7}.9401 \approx 0.1259$; $x = 2.01$: $1/(2.01 \cdot 4.01) = \frac{1}{8}.0601 \approx 0.1241$ — $0.125 = \frac{1}{8}$ ga yaqinlashadi. Algebra jadvalni tasdiqlaydi: limit $1/(2 \cdot 4) = \frac{1}{8}$. T Jadvaldan taxmin qiling, keyin grafik bilan tekshiring: $\lim_{\theta\to 0} \sin(\pi/\theta)$.

[Manba: OpenStax §2.2, mashq 44]

Yechim. Qiymatlar: $\theta = 0.1$: $\sin(10\pi) = 0$; 0.01: $\sin(100\pi) = 0$; 0.001: $\sin(1000\pi) = 0$; 0.0001: 0. Jadval 0 ga intiladi — lekin $\sin(\pi/\theta)0$ ga yaqin $-1va1$ oralig‘ida cheksiz tez tebranadi (masalan, $\theta = \frac{2}{3}\sin(3\pi/2) = -1$ beradi, $\theta = \frac{2}{5}esa\sin(5\pi/2) = 1$). Limit mavjud emas; jadval aldaydi, chunki tanlangan nuqtalar $1/n$ — sinusning aniq

nollari. Bu jadval usuliga oid ogohlantirish mashqi. T Jadvaldan taxmin qiling, keyin grafik bilan tekshiring: $\lim_{\alpha\to 0+}(1/\alpha)\cdot\cos(\pi/\alpha)$.

[Manba: OpenStax §2.2, mashq 45]

Yechim. $\alpha = 1/n$ bo‘lganda: $(1/\alpha)\cdot\cos(n\pi) = n\cdot(-1)^n$ — $\pm n$ qiymatlari modulda o‘sadi va belgilar almashib turadi. Grafik tobora katta musbat va manfiy qiymatlar oralig‘ida tebranadi: limit mavjud emas ($na+\infty$, na $-\infty$ — belgilar almashadi). Jadval omadli nuqtalarda ($\alpha = 1/(2n) + 2n \to +\infty$ beradi) qarama-qarshi tarafda aldaydi.